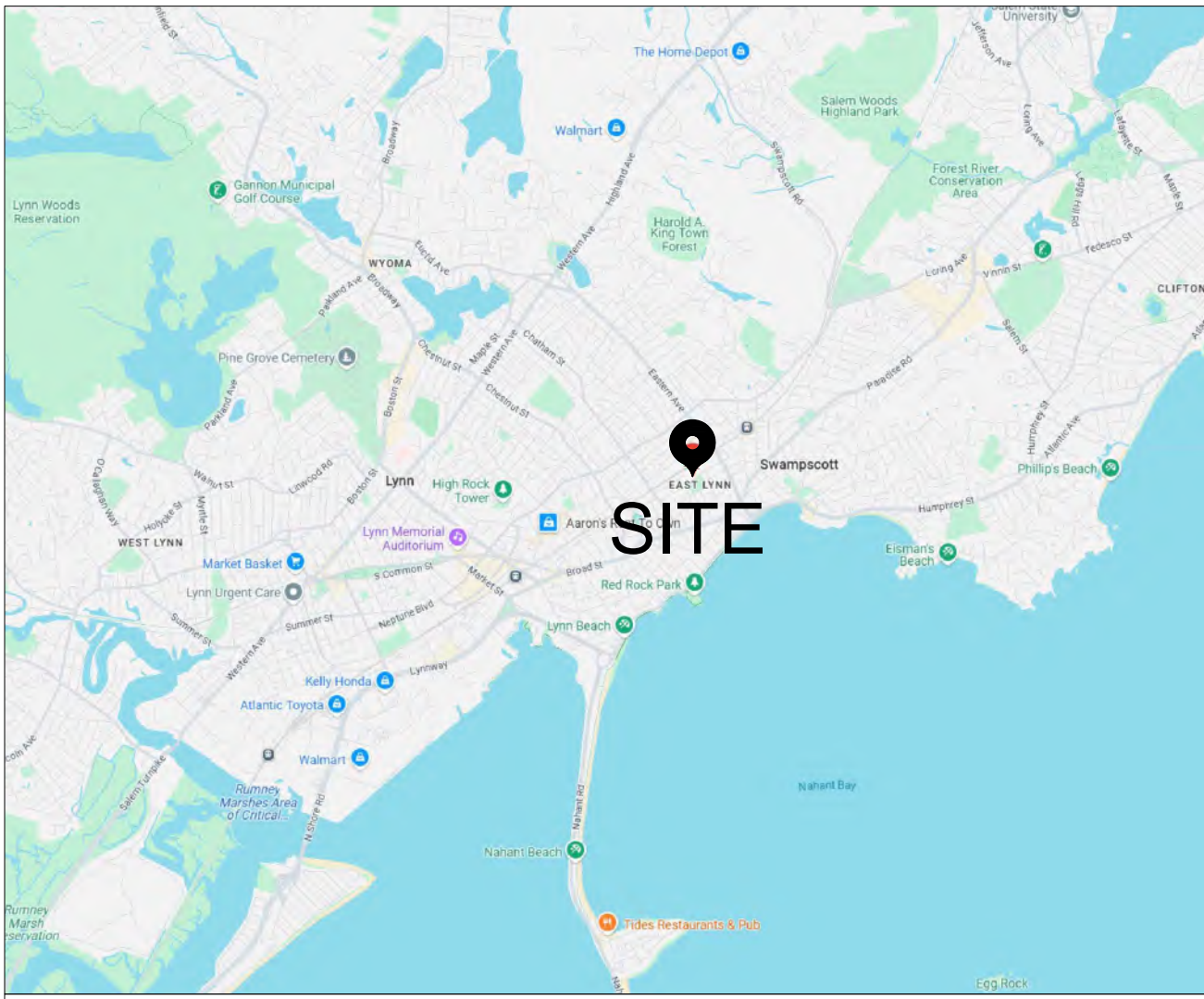




LOCATION MAP

SITE COORDINATES	
LATITUDE:	42°28'13.57"N
LONGITUDE:	70°55'41.14"W

ASHRAE TEMPERATURES			
BEVERLY, MA, USA (WMO: 725088)			
ELEV.	HIGH TEMPERATURE		LOW TEMPERATURE
	0.4%	2.0%	EXTREME MINIMUM
33m	34.1°C	31.9°C	-27.9°C

DESIGN LOADS	
GROUND SNOW LOAD	40 PSF
FLAT ROOF SNOW LOAD	40 PSF
WIND LOAD	119 MPH

PROJECT DATA
CODES: NEC 2023 (NFPA 70) INCLUDING AMENDMENTS AND MODIFICATIONS MADE BY THE BFPR AND DULY PROMULGATED IN 527 CMR 12.00 MA BUILDING CODE NINTH EDITION CMR 780

SYSTEM DESCRIPTION:
SOLAR ARRAYS: DC OUTPUT (STC): 248.22 (±) kW DC AC OUTPUT: 180.00 kW AC NOMINAL 180.00 kW AC MAXIMUM
EQUIPMENT: PANELS: (394) HANWHA Q CELLS Q.TRON XL-G2.3 / BFG 630W PV MODULE INVERTERS: (1) SOLIS S5-GC100K-US 100.00kW (2) SOLIS S6-GC40K-US 40.00kW
RSD's: (45) APSmart RSD-D DUAL MODULE RAPID SHUTDOWN DEVICES
RACKING (FLAT ROOF): K2's THE SOUTH FACE SYSTEM 10° FLAT ROOF RACKING UL3741 COMPLIANT RACKING SYSTEM (OR SIMILAR)
RACKING (ELEVATED RACKING): ISA CORP WSS ELEVATED RACKING SYSTEM UL3741 COMPLIANT RACKING(OR SIMILAR)

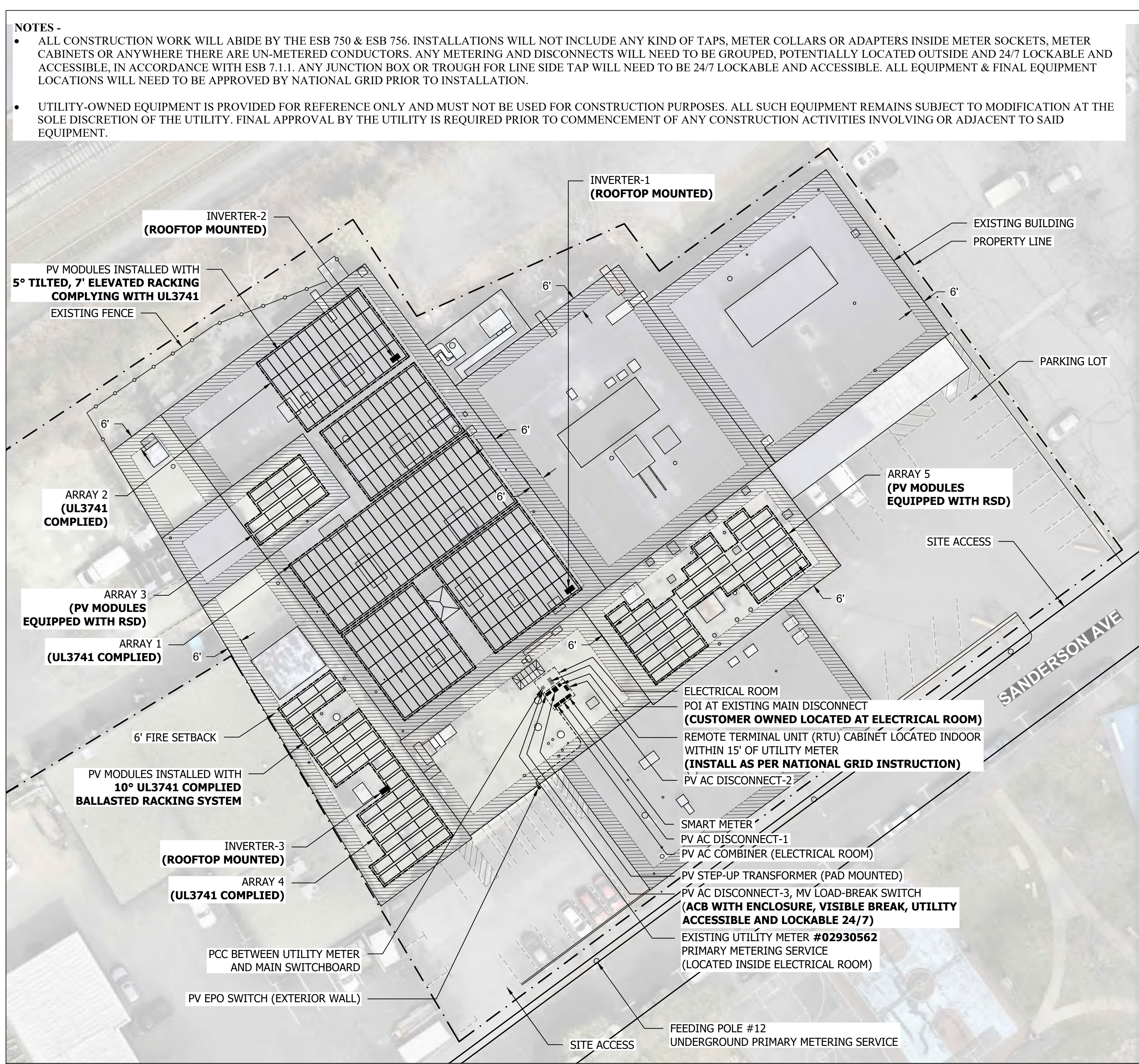
DWARKA LLC

86 SANDERSON AVENUE, LYNN, MA 01902

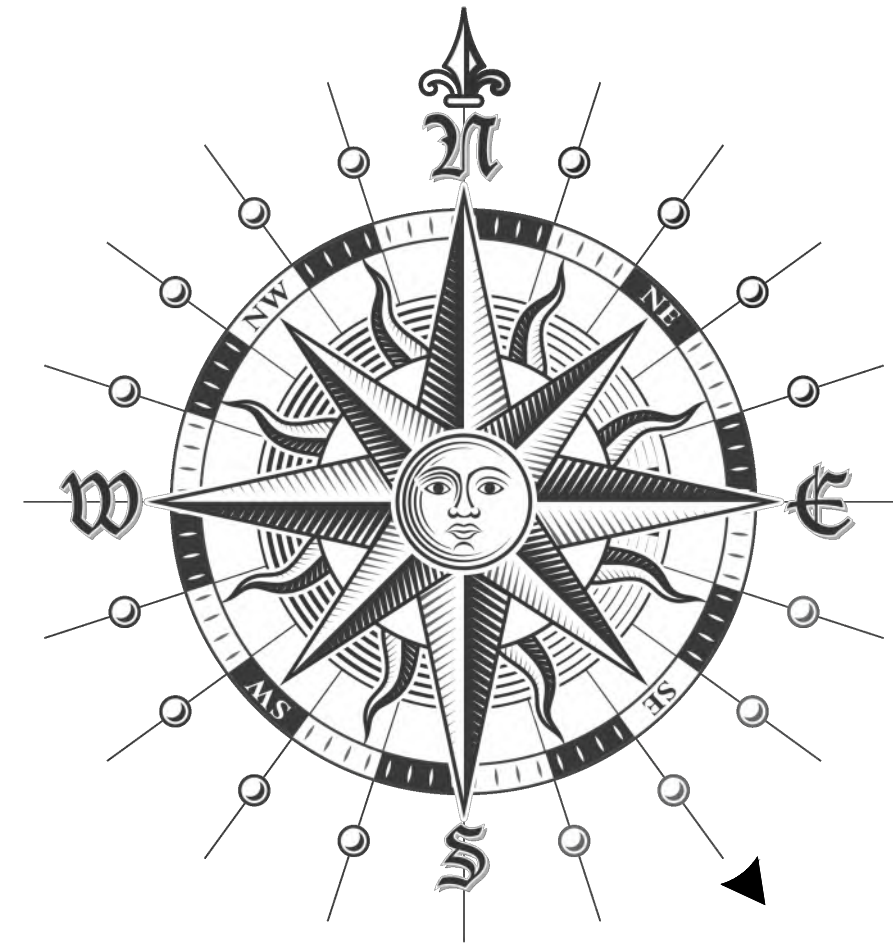
BTM ROOFTOP PHOTOVOLTAIC SYSTEM

248.22 KW-DC 180.00 KW-AC

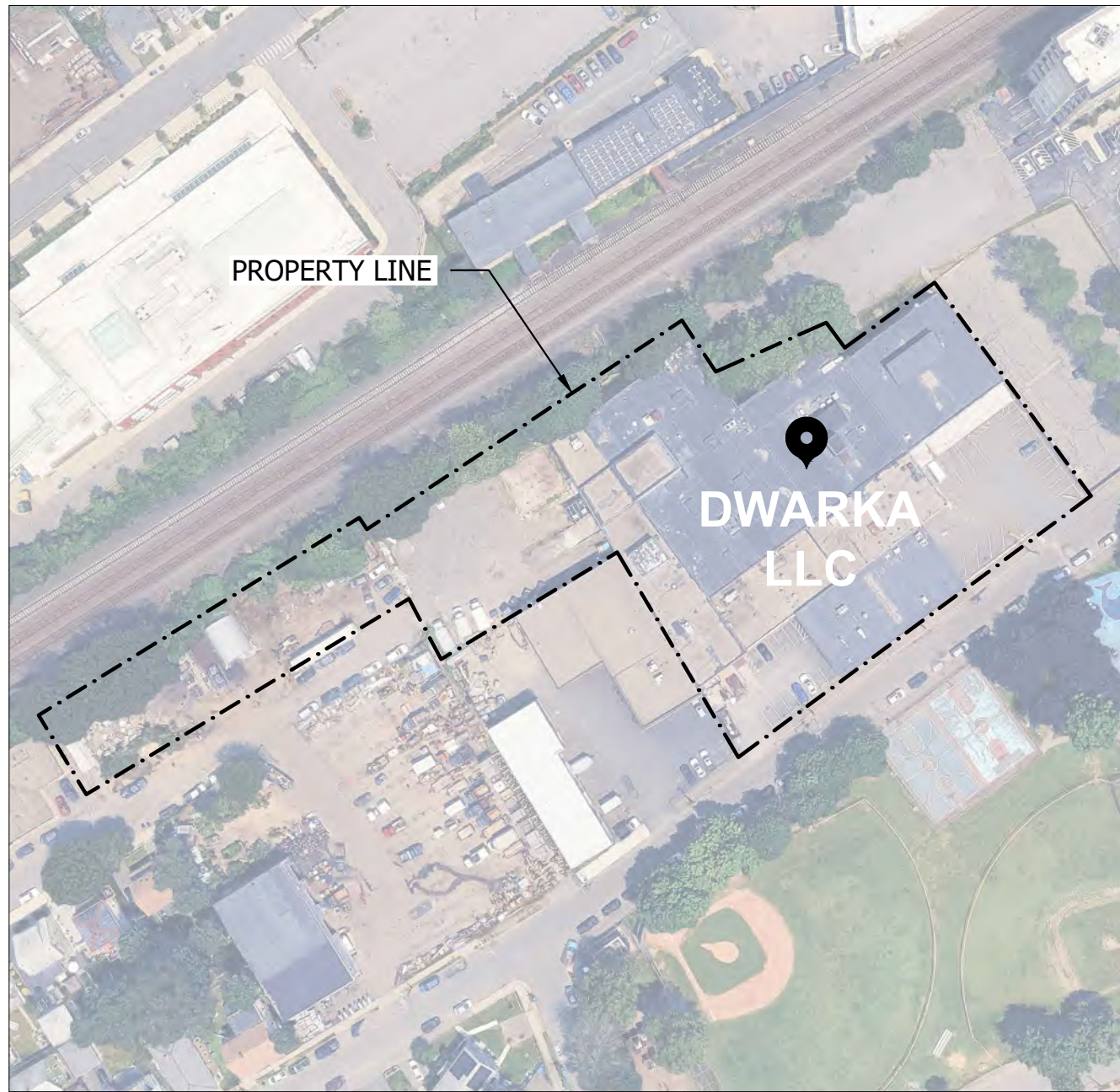
FOR PERMITTING PURPOSES ONLY



SITE PLAN



ROOFTOP
MODULES FACING
SE - 143° TILT - 10°
SE - 143° TILT - 5°



SITE OVERVIEW

SHEET LIST	
A-101:	SITE PLAN, LOCATION AND PROJECT OVERVIEW
A-102:	ENLARGED SITE PLAN
A-103:	K2 RACKING UL 3741 CERTIFICATE AND SPECS
A-104:	ISA RACKING UL CERTIFICATE AND SPECS
E-101:	SINGLE LINE DIAGRAM
E-102:	WIRING DETAILS AND CALCULATIONS
E-103:	DATA ACQUISITION SYSTEM
E-201:	LABELS AND PLACARDS
E-202:	INVERTER AND MODULE SPECIFICATIONS
E-203:	RACKING AND RSD SPECIFICATION
E-204:	AC COMBINER SPECIFICATION
E-205:	TRANSFORMER SPECIFICATION
E-206:	PV MV DISCONNECT SPECIFICATION
E-207:	GENERAL AND ELECTRICAL NOTES
K2 RACKING PLANSET (ATTACHED SEPARATELY)	
K2 STRUCTURAL ANALYSIS REPORT (ATTACHED SEPARATELY)	
ISA RACKING PLANSET (ATTACHED SEPARATELY)	
ISA STRUCTURAL ANALYSIS REPORT (ATTACHED SEPARATELY)	

PROJECT DESIGN

SURYA
Standard

Harnessing the best source of energy available

3 Allied Drive, Suite 107
Dedham, MA 02026
amanullah@suryastandard.com
(617) 842-8016

SEAL/SIGNATURE

sealed 03aug2026 mike@h2dc.com
H2DC PLLC MA CoA#: 001239603
ELECTRICAL ONLY
FOR PERMITTING PURPOSES ONLY

PROJECT

BTM ROOFTOP
PHOTOVOLTAIC SYSTEM

DWARKA LLC
86 SANDERSON AVENUE, LYNN, MA 01902

PROJECT OWNER
DWARKA LLC

PROJECT DEVELOPER

SURYA
Standard

Harnessing the best source of energy available

3 Allied Drive, Suite 107
Dedham, MA 02026
amanullah@suryastandard.com
(617) 842-8016

REVISIONS		
DATE	COMMENT	REV.
5/11/2026	INTERCONNECTION PLAN SET	1.0
8/3/2026	PERMIT PLAN SET	1.0

PROJECT NO:
FILE NAME: DWARKA LLC
DRAWN BY: SPV
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PERMIT PLAN SET

SHEET TITLE
SITE PLAN, LOCATION AND
PROJECT OVERVIEW

SHEET NUMBER
A-101

LEGENDS

NEW PV MODULE

NEW ELECTRICAL EQUIPMENT

EXISTING ELECTRICAL EQUIPMENT

NEW PV CONDUIT

PROPERTY LINE

FIRE SETBACK

STR #

STRING LABEL

11 MODULE STRING

13 MODULE STRING

14 MODULE STRING

15 MODULE STRING

1

ROOF NUMBER LABEL

VENT PIPE

CHIMNEY

ARRAY INFO

A1

AZIMUTH - 143°
PANEL TILT- 5°
MODULE QTY - 240

A2

AZIMUTH - 143°
PANEL TILT- 5°
MODULE QTY - 52

A3

AZIMUTH - 143°
PANEL TILT- 10°
MODULE QTY - 14

A4

AZIMUTH - 143°
PANEL TILT- 10°
MODULE QTY - 43

A5

AZIMUTH - 143°
PANEL TILT- 10°
MODULE QTY - 45

NOTE - ALL SOLAR PV EQUIPMENT, INCLUDING PV MODULES, RACKING COMPONENTS, AND ASSOCIATED HARDWARE, SHALL BE INSTALLED IN ACCORDANCE WITH THE APPLICABLE MANUFACTURER'S INSTALLATION INSTRUCTIONS AND GUIDELINES.

NOTE 1 - EXISTING ROOF CONSTRUCTION CONSISTS OF EPDM RUBBER MEMBRANE ROOFING OVER A SOLID CONCRETE STRUCTURAL DECK. ALL PROPOSED WORK SHALL BE COORDINATED WITH THE EXISTING ROOF ASSEMBLY.

NOTE 2 - ALL ELECTRICAL EQUIPMENT SUCH AS PV COMBINER, PV DISCONNECT -1, PV DISCONNECT -2, PV STEP-DOWN TRANSFORMER AND 5 KV LOAD INTERRUPTER SWITCH WILL BE INSTALL AT BASEMENT INSIDE THE EQUIPMENT ROOM BESIDE THE EXISTING ELECTRICAL ROOM, INSTALLER WILL INSTALL A SMOKE DETECTORS AND LINK WITH EXISTING BUILDING FIRE ALARM SYSTEM.

EXISTING BUILDING PROPERTY LINE

6' FIRE SETBACK

PV CONDUIT

PV EPO SWITCH (EXTERIOR WALL)

PV AC DISCONNECT-3
MV LOAD-BREAK SWITCH
(ACB WITH ENCLOSURE, VISIBLE BREAK,
UTILITY ACCESSIBLE AND LOCKABLE 24/7)

PV STEP-UP TRANSFORMER
(PAD MOUNTED)

PV AC COMBINER (ELECTRICAL ROOM)

PV AC DISCONNECT-1
SMART METER

FEEDING POLE #12
UNDERGROUND PRIMARY METERING SERVICE

PCC BETWEEN UTILITY METER
AND MAIN SWITCHBOARD
EXISTING UTILITY METER
#02930562 PRIMARY
METERING SERVICE
(LOCATED INSIDE
ELECTRICAL ROOM)

ELECTRICAL ROOM
POI AT EXISTING MAIN DISCONNECT
(CUSTOMER OWNED LOCATED AT ELECTRICAL ROOM)
REMOTE TERMINAL UNIT (RTU) CABINET LOCATED INDOOR
WITHIN 15' OF UTILITY METER
(INSTALL AS PER NATIONAL GRID INSTRUCTION)
PV AC DISCONNECT-2

EXISTING 8'X10' EQUIPMENT ROOM BESIDE
ELECTRICAL ROOM ALL EQUIPMENT WILL BE
LOCATED INSIDE THIS ROOM, NEED TO
INSTALL SMOKE DETECTORS

INVERTER-1
(ROOFTOP MOUNTED)

INVERTER-2
(ROOFTOP MOUNTED)

INVERTER-3
(ROOFTOP MOUNTED)

ARRAY 1
(UL3741 COMPLIED)

ARRAY 2
(UL3741 COMPLIED)

ARRAY 3
(14 PV MODULES
EQUIPPED WITH RSD)

ARRAY 4
(UL3741 COMPLIED)

ARRAY 5
(45 PV MODULES
EQUIPPED WITH RSD)

(43) HANWHA Q CELLS Q.TRON XL-G2.3
/ BFG 630W
PV MODULES INSTALLED WITH
10° UL3741 COMPLIED BALLASTED
RACKING SYSTEM

(240) HANWHA Q CELLS Q.TRON XL-G2.3
/ BFG 630W PV MODULES INSTALLED
WITH
7° TILTED, 7' ELEVATED RACKING
COMPLYING WITH UL3741

(45) APSMART RSD-D DUAL MODULE RSD
(ONE PER TWO MODULE)

PARKING LOT

SANDERSON AVE

SITE PLAN

FOR PERMITTING PURPOSES ONLY

PROJECT DESIGN

3 Allied Drive, Suite 107
Dedham, MA 02026
amanullah@suryastandard.com
(617) 842-8016

SEAL/SIGNATURE

sealed 03aug2026 mike@h2dc.com
H2DC PLLC MA CoA#: 001239603
ELECTRICAL ONLY
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PROJECT

BTM ROOFTOP PHOTOVOLTAIC SYSTEM

DWARKA LLC
86 SANDERSON AVENUE, LYNN, MA 01902

PROJECT OWNER
DWARKA LLC

PROJECT DEVELOPER

3 Allied Drive, Suite 107
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amanullah@suryastandard.com
(617) 842-8016

REVISIONS

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5/11/2026	INTERCONNECTION PLAN SET	1.0
8/3/2026	PERMIT PLAN SET	1.0

PROJECT NO:

FILE NAME: DWARKA LLC

DRAWN BY: SPV

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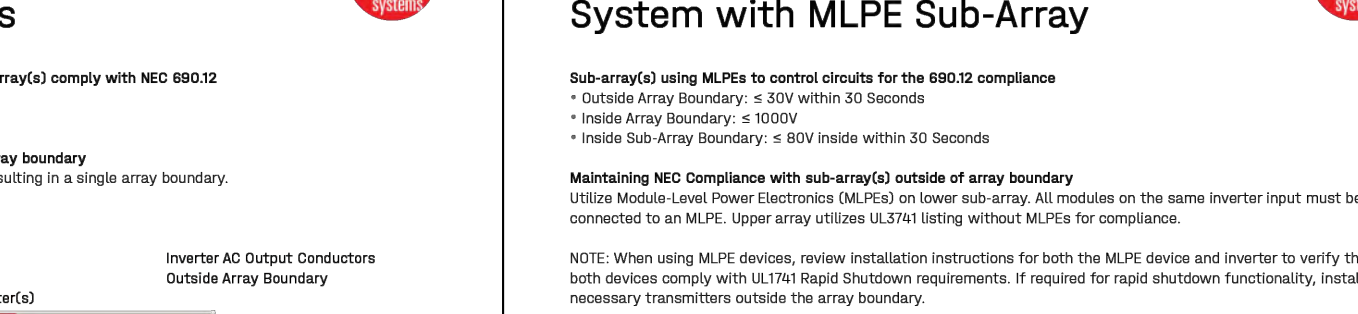
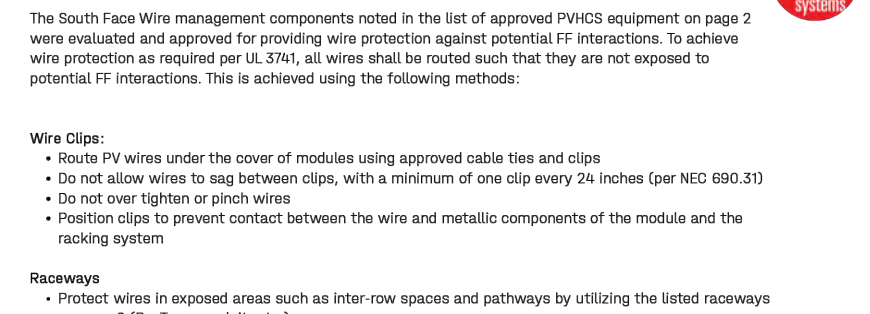
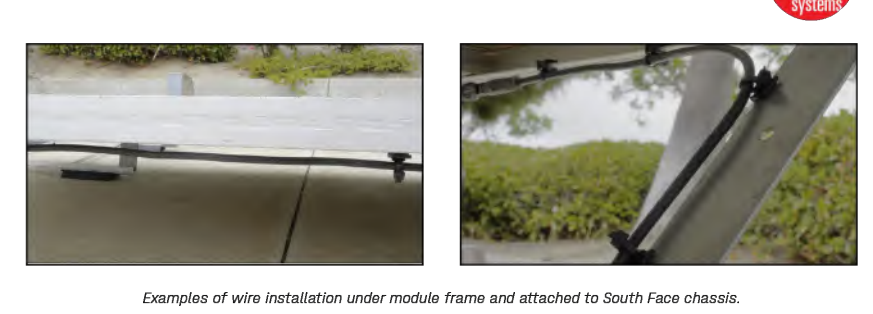
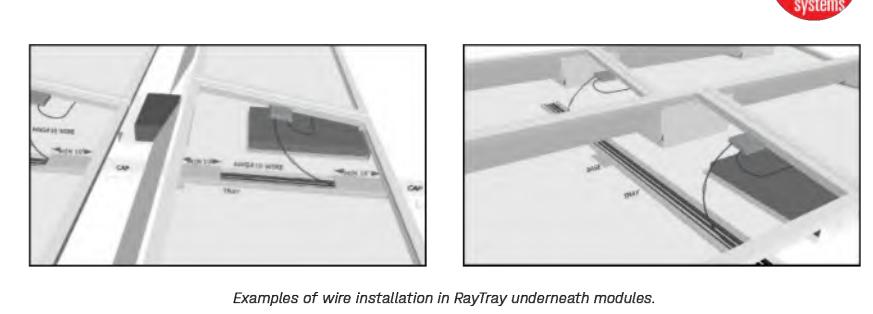
PERMIT PLAN SET

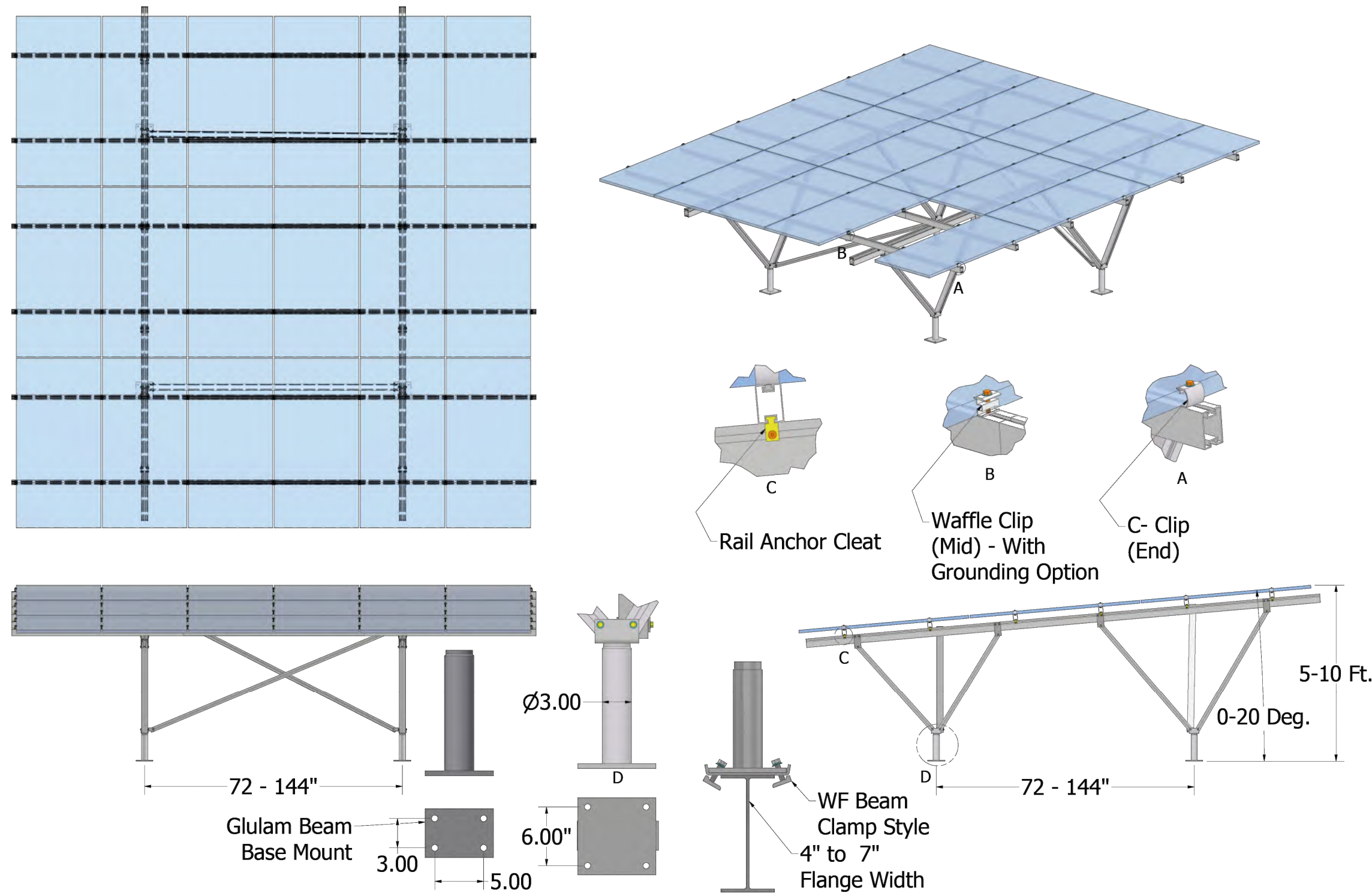
SHEET TITLE

SITE LAYOUT AND STRING PLAN

SHEET NUMBER

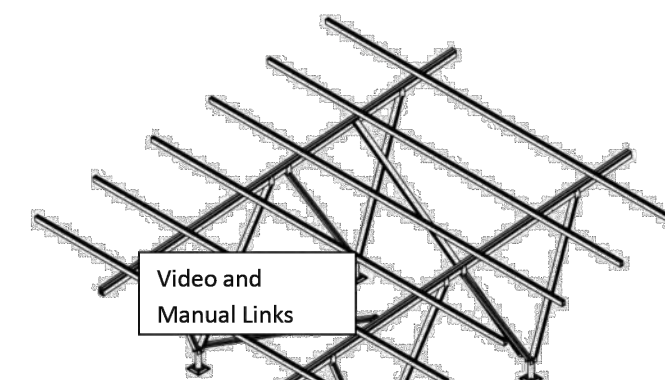
A-102

Case 2: UL 3741 Listed System with Contiguous Sub-Arrays Sub-Array(s) are within the same Array Boundary and Array(s) comply with NEC 690.12 - Outside Array Boundary: ≤ 50V within 30 Seconds - Inside Array Boundary: ≤ 1500V - Inside Sub-Array Boundary: ≤ 60V inside within 30 Seconds Maintaining NEC Compliance with sub-array(s) within array boundary: Maximum 2 ft. spacing between all array components resulting in a single array boundary.  The diagram shows a rectangular array of solar panels divided into two sections by a vertical gap. Labels indicate 'Inverter DC Input Conductors Within Array Boundary' on the left, 'Inverter(s)' in the center gap, and 'Inverter AC Output Conductors Outside Array Boundary' on the right. Dimensions include 'Array Boundary 1 ft.', 'Required Setback (varies)', 'Series String Wiring', 'Obstruction Affects Array Layout', and 'Array < 160 ft.'. A callout shows a close-up of the gap with the dimension '≤ 2 ft.' and a note: 'All modules within same array boundary'.	Case 3: South Face UL 3741 Listed System with MLPE Sub-Array Sub-Array(s) using MLPEs to control circuits for the 690.12 compliance - Outside Array Boundary: ≤ 50V within 30 Seconds - Inside Array Boundary: ≤ 1500V - Inside Sub-Array Boundary: ≤ 60V inside within 30 Seconds Maintaining NEC Compliance with sub-array(s) outside of array boundary Utilize Module-Level Power Electronics (MLPE) on lower sub-array. All modules on the same inverter input must be connected to an MLPE. Upper array utilizes UL1741 listing without MLPEs for compliance. NOTE: When using MLPE devices, review installation instructions for both the MLPE device and inverter to verify that both devices comply with UL1741 Rapid Shutdown requirements. If required for rapid shutdown functionality, install necessary transmitters outside the array boundary.  The diagram shows a rectangular array of solar panels. The bottom section is labeled 'Sub-Array'. Labels include 'Inverter DC Input Conductors Within Array Boundary', 'Inverter(s)', 'Inverter AC Output Conductors Outside Array Boundary', 'Array Boundary 1 ft.', 'Required Setback (varies)', 'Series String Wiring', 'Array Dimension >150 ft. Requires Access Pathway Pathways Creates Two Distinct Array Boundaries', and 'Array < 160 ft.'. A callout shows a close-up of the bottom section with the number '3.' and a note: 'Listed Module-Level Power Electronics'.	UL 3741 Wire Management Guidelines The South Face Wire management components noted in the list of approved PIVCS equipment on page 2 were evaluated and approved for providing wire protection against potential FF interactions. To achieve wire protection as required per 0.3NA, all wires shall be routed such that they are not exposed to potential FF interactions. This is achieved using the following methods: Wire Clips: - Rackle PV wires under the cover of modules using approved cable tie and clips - Do not allow wires to sag between clips, with a minimum of one clip every 24 inches (per NEC 690.31) - Do not over tighten or pinch wires - Pullion clips to prevent contact between the wire and metallic components of the module and the racking system Raceways: - Protect wires in exposed areas such as inter-row spaces and pathways by utilizing the listed raceways on page 2 (Baylayr, conduit, etc.) - Ensure that PV wire is not exposed to sharp edges when entering or exiting the raceway See photos for examples of end module and under chassis wire management  The four photos show: 1) Wire clips being used to secure wires under a solar panel. 2) Wires running through a white plastic raceway. 3) Wires being managed at the end of a solar panel array. 4) Wires being protected under a solar panel chassis.	UL 3741 Wire Management Guidelines Manage larger bundles of cables using RayRay® or another listed raceway. When entering and exiting the raceway, installers shall ensure PV wires are routed away from exposure to Fire Fighter interactions.  The two photos show: 1) A bundle of wires being inserted into a black RayRay raceway. 2) A bundle of wires secured within a RayRay raceway using a strap.
--	--	---	--



Union City CA 94587
510-324-3755

- Allows Edge to Edge Modules on Roof to Maximize Solar
- Clears AC & Roof Equipment
- Simplifies re-Roofing
- Fire Access underneath Array
- Tilt can be field adjusted



- Tilt Angle: 0-10 Degrees Standard
- Anchoring: Flanged Base & Clamped for Glulam & Steel, Clamped for Service Stations
- Materials: 6000 Series Aluminum with 300 Series Stainless Fasteners.
- Weight: 1 Lb. per square foot
- Warranty: 20 Years
- 2703 Certified
- Manufactured in USA

Union City CA 94587
info@isa-corporation.com

	JAP6(K)-72-xxx/48B, JAP72SY-xxx/ZZ, JAP6(K)-60-xxx/48B, JAP60SY-xxx/ZZ, JAM6(K)-72-xxx/ZZ, JAM72SY-xxx/ZZ, JAM6(K)-60-xxx/ZZ, JAM60SY-xxx/ZZ, 1. Y: 01, 02, 03, 09, 10 2. Z: SC, PR, BP, HT, IB, MW YY = Backsheet, ZD Cell technology JAM72D30-xxx/MB, JAM54531-xxx/GR, JAM54530-xxx/MB, JAM54531-xxx/GR
Jinko Solar	JKM & JKMS Series JKMxxxM-72HL-V JKMxxxM-54HL4-B, JKMxxxM-78L3-TV JKMxxxM-6R13-B, JAMxxxM-72HL-TV JKMxxxM-72HL4-TV, JKMxxxM-72HL-B JKMxxxM-72HL-V-BOW, JKMxxxM-72HL4-BDV.30IK JKMxxxM-72HL4-BDV.30MC, JKMxxxM-72HL4-BDV.35IK JKMxxxM-72HL4-BDV.35MC
LG	LG425QA-A6
Longi	LR4-60(HPB/HPH), LR4-72(HBD/HPH), LR6-60, LR6-60(B)/HPB/HPH/PB/PE/PH), LR6-72, LR6-72(B)/HB/HPB/PE/PH), RealBlack LR4-60HPB, RealBlack LR6-60HPB, LR5-72HPH, LR5-72HPH-xxxM LR5-54HAB-xxxM, LR5-72HBD-xxxM LR4-60HPB-xxxM, LR5-54HAB-xxxM-V5
Maxeon	SPR-MAX3-XXX-R SPR-MAX-xxx-BLK-R, SPR-MAX6-425 BLK SPR-MAX6-xxx, SPR-MAX6-xxx-CON, SPR-MAX3-xxx-BLK-R
Meyer Burger	MB_B120AYB-XXX, MB_B108CYC_XXX MB-TG120BYB-xxx, MB-W120AYB-xxx
Mission Solar Energy	MSE Mono, MSE Perc MSE415SX62, MSExxxSX5R MSExxxSX6W, MSExxxSX6M MSExxxSX9R, MSExxxSX9Z
Mitsubishi	MJE & MLE Series
Neo Solar Power Co.	D6M Series
Panasonic	VBHNxxxSA06/SA06B/SA11/SA11B, VBHNxxxSA15/SA15B/SA16/SA16B, VBHNxxxKA, VBHNxxxKA03/04, VBHNxxxSA17/SA17G/SA17E/SA18/SA18E, VBHNxxxZA01/ZA02/ZA03/VBHNxxxZA04 EVPVxxx, EVPVxxx(H/K/PK/HK/HK2) EVPVxxxPK-2
Peimar	S6xxxM (FB/BF), S6xxxM
Phono Solar	PSxxxM1H-20/U, PSxxxM1-20UH, PSxxxM1-20UH

30

Attention: Tony Zante



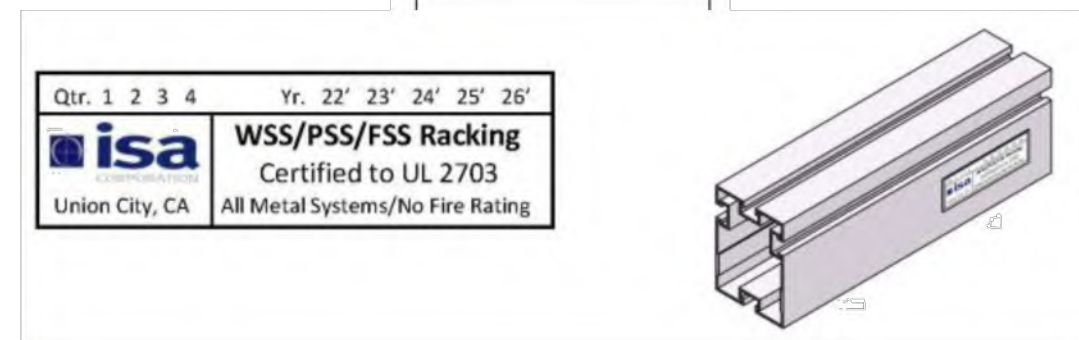
PRODUCTS
CLASS - C531382 - POWER SUPPLIES - PHOTOVOLTAICS-PV Racking and clamping systems -
Certified to US Standards

Models:	WSS	- WSS Elevated Racking System, PV racking system for bonding and grounding of photovoltaic modules
	PSS	- PSS Elevated Racking System, PV racking system for bonding and grounding of photovoltaic modules
	FSS	- FSS Flush Mount Racking System, PV racking system for bonding and grounding of photovoltaic modules

Page 1

1. Submitter's name and/or CSA Master Contract number "301533".
2. Model designation.
3. Manufacturing date, year-quarter;
4. "Not Fire Rated"

Page 2



Page 3

Project	Date	Description
80099468	2022-06-07	Evaluation, testing, and certification of models WSS, PSS, and FSS. Solar Photovoltaic Mounting System for rooftop applications for CSAus. Will not be fire rated.

Page 1

SEAL/SIGNATURE

DWARKA LLC
86 SANDERSON AVENUE, LYNN, MA 01902

PROJECT OWNER
DWARKA LLC

SURYA
Standard
Harnessing the best source of energy available

3 Allied Drive, Suite 107
Dedham, MA 02026
anullah@suryastandard.com
(617) 842-8016

DATE	COMMENT	REV
5/11/2026	INTERCONNECTION PLAN SET	1.0
8/3/2026	PERMIT PLAN SET	1.0

PROJECT NO:

FILE NAME: DWARKA LLC

DRAWN BY: SPV

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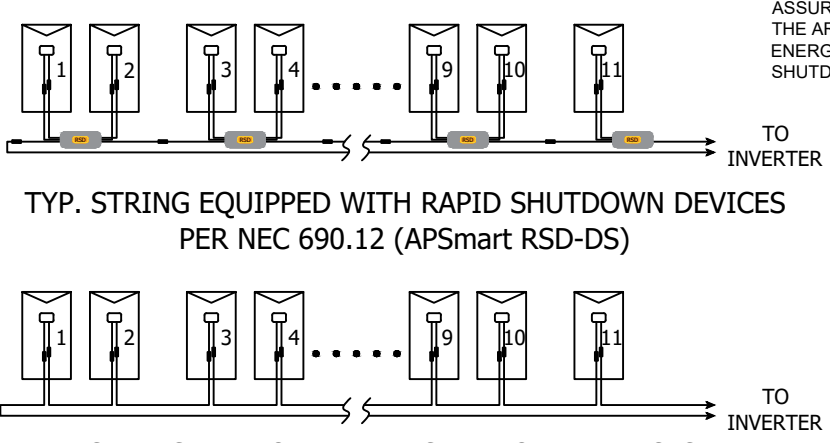
SHEET TITLE
**ISA CORP. WSS-6
RACKING SPECIFICATIONS**

SHEET NUMBER

A-104

FOR PERMITTING PURPOSES ONLY

DC OUTPUT (STC): 248.22 (±) kW DC
AC OUTPUT: 180.00 kW AC NOMINAL
180.00 kVA AC MAXIMUM
DC-AC RATIO: 1.38



Inverter Protective Settings ISO-NE SRD Tables I & II				
INTERNAL PROTECTIVE FUNCTIONS	TRIP OUTPUT	VOLTAGE SETTING PU	SETTING	CLEARING TIME (s)
27-1 UNDERVOLTAGE	X	0.5	138.5 V	1.1
27-2 UNDERVOLTAGE	X	0.88	243.76 V	3
59-1 OVERVOLTAGE	X	1.1	304.7 V	2
59-2 OVERVOLTAGE	X	1.2	332.4 V	0.16
81U-1 UNDERFREQUENCY	X		56.5 Hz	0.16
81U-2 UNDERFREQUENCY	X		58.5 Hz	300
81O-1 OVERFREQUENCY	X		61.2 Hz	300
81O-2 OVERFREQUENCY	X		62.0 Hz	0.16
PRIMARY VOLTAGE (L-N)	0.277	kV	277	V

NOTES

1.ALL CONSTRUCTION WORK WILL ABIDE BY THE ESB 750 & ESB 756. INSTALLATIONS WILL NOT INCLUDE ANY KIND OF TAPS, METER COLLARS OR ADAPTERS INSIDE METER SOCKETS, METER CABINETS OR ANYWHERE THERE ARE UN-METERED CONDUCTORS. ANY METERING AND DISCONNECTS WILL NEED TO BE GROUPED, POTENTIALLY LOCATED OUTSIDE AND 24/7 LOCKABLE AND ACCESSIBLE, IN ACCORDANCE WITH ESB 7.1.1. ANY JUNCTION BOX OR TROUGH FOR LINE SIDE TAP WILL NEED TO BE 24/7 LOCKABLE AND ACCESSIBLE. ALL EQUIPMENT & FINAL EQUIPMENT LOCATIONS WILL NEED TO BE APPROVED BY NATIONAL GRID PRIOR TO INSTALLATION.

WIRING SCHEDULE (ALL CONDUCTORS ARE COPPER XHHW-2 UNLESS OTHERWISE NOTED)									
DC HOME RUNS - FROM MODULES AT BEGINNING AND END OF STRINGS TO INVERTERS									
	WIRE TYPE	COND. SIZE	GND		RACEWAY		DC LOAD	DC OVERCURRENT	LENGTHS
A	SOLAR PV WIRE	#10	#6 BARE Cu		EMT < 1" OR RAYTRAY		13.88 A	17.35 A / 21.68 A	15'-200'
INVERTERS TO PV AC COMBINER									
	L1, L2, L3	N	GND	CONDUIT	NO. SETS	AC LOAD	AC OVERCURRENT	OCPD SIZE	LENGTHS
B1	#2/0 Cu (3)	N/A	#6 Cu	2" EMT	1	120.3 A	150.375 A	175 A	INV 1 - 150'
B2	#4 Cu (3)	N/A	#8 Cu	2" EMT	1	48.1 A	60.125 A	70 A	INV 2 - 250', INV 3 - 50'
AC COMBINER TO PV DISCONNECT -1/ PV DISCONNECT -1 TO SMART METER/ SMART METER TO PV DISC - 2/PV DISCONNECT - 2 TO PV TRANSFORMER									
	L1, L2, L3	N	GND	CONDUIT	NO. SETS	AC LOAD	AC OVERCURRENT	OCPD SIZE	LENGTHS
C	#1/0 Cu (3)	#1/0 Cu	#4 Cu	2" EMT	2	216.50A	270.625 A	300 A	INV TO PV XFMR - 70'
PV TRANSFORMER TO PV DISCONNECT NO. 3 / PV DISCONNECT NO. 3 TO SWITCHBOARD									
	L1, L2, L3	N	GND	CONDUIT	NO. SETS	AC LOAD	AC OVERCURRENT	OCPD SIZE	LENGTHS
D	#6 Cu (3) MV XLPE CABLE	#6 Cu MV XLPE CABLE	#6 Cu	1.5" EMT	1	24.98 A	31.225 A	40 A	PV XFMR TO SMART METER - SMART METER TO PV DISCO1 PV DISCO1 TO PV DISCO2 - 40' PV DISCO2 TO POI - 10'

WIRING SCHEDULE (ALL CONDUCTORS ARE COPPER XHHW-2 UNLESS OTHERWISE NOTED)									
DC HOME RUNS - FROM MODULES AT BEGINNING AND END OF STRINGS TO INVERTERS									
	WIRE TYPE	COND. SIZE	GND		RACEWAY		DC LOAD	DC OVERCURRENT	LENGTHS
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INVERTERS TO PV AC COMBINER									
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B1	#2/0 Cu (3)	N/A	#6 Cu	2" EMT	1	120.3 A	150.375 A	175 A	INV 1 - 150'
B2	#4 Cu (3)	N/A	#8 Cu	2" EMT	1	48.1 A	60.125 A	70 A	INV 2 - 250', INV 3 - 50'
AC COMBINER TO PV DISCONNECT -1/ PV DISCONNECT -1 TO SMART METER/ SMART METER TO PV DISC - 2/PV DISCONNECT - 2 TO PV TRANSFORMER									
	L1, L2, L3	N	GND	CONDUIT	NO. SETS	AC LOAD	AC OVERCURRENT	OCPD SIZE	LENGTHS
C	#1/0 Cu (3)	#1/0 Cu	#4 Cu	2" EMT	2	216.50A	270.625 A	300 A	INV TO PV XFMR - 70'
PV TRANSFORMER TO PV DISCONNECT NO. 3 / PV DISCONNECT NO. 3 TO SWITCHBOARD									
	L1, L2, L3	N	GND	CONDUIT	NO. SETS	AC LOAD	AC OVERCURRENT	OCPD SIZE	LENGTHS
D	#6 Cu (3) MV XLPE CABLE	#6 Cu MV XLPE CABLE	#6 Cu	1.5" EMT	1	24.98 A	31.225 A	40 A	PV XFMR TO SMART METER - SMART METER TO PV DISCO1 PV DISCO1 TO PV DISCO2 - 40' PV DISCO2 TO POI - 10'

CONSTRUCTION / NEC NOTES		
INTERCONNECTION TO UTILITY AND SYSTEM GROUNDING PER NEC-2020 ARTICLE 690.	PROVIDE SIGNAGE AS REQ'D BY NEC-2020 ARTICLE 690. ALL OUTDOOR EQUIPMENT SHALL BE A MINIMUM OF NEMA-3R RATED.	AC & DC GROUNDING CONDUCTORS PER NEC ARTICLE 690.43(A) THRU (F) CONNECTED AS PER 690.45(A), 690.46, & SIZED PER 250.122.
*NOTE: THE PV DISCONNECT IS TO BE INSTALLED ON THE EXTERIOR WALL OF THE ELECTRICAL ROOM, BACK-TO-BACK WITH THE MAIN SWITCHBOARD, WHERE THE POI IS LOCATED. THUS, THE CONDUCTOR LENGTH BETWEEN THE DISCONNECT AND THE LINE-SIDE TAP (POI) IS ONLY ON THE ORDER OF 5-7'.		



PROJECT DESIGN

SURYA
Standard

Harnessing the best source of energy available

3 Allied Drive, Suite 107
Dedham, MA 02026
amanullah@suryastandard.com
(617) 842-8016



MAICHAEL A.
McGUIRE
ELECTRICAL
NO. 38615
REGISTERED
PROFESSIONAL ENGINEER

Sealed 03aug2026 mike@gh2dc.com
H2DC PLLC MA CoA#: 001239603
ELECTRICAL ONLY
FOR PERMITTING PURPOSES ONLY

PROJECT DEVELOPER



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SHEET NUMBER

E-101

AC WIRING DETAILS & VOLTAGE DROP CALCULATIONS FOR AC WIRING BETWEEN INVERTERS AND POI

Inv No.	Phase Conductors (XHHW-2)				Grounded Conductor/Neutral (XHHW-2)			Equipment Grounding Conductor (XHHW-2)		Conduit Between Inverters and AC Combiner				Parameters for Voltage Drop Calculation								Approximate (APP) EVD = $\sqrt{3}/1000 \times (IRL\cos\theta + IXL\sin\theta)$ ($IRL\cos\theta + IXL\sin\theta$)			Exact (EX) EVD = $\sqrt{3} \times \{ES + (IRL\cos\theta + IXL\sin\theta)/1000 - \sqrt{[VS2 - ((IXL\cos\theta - IRL\sin\theta)/1000)^2]}\}$		
	L-L-L	Conduit Material	Area (mm²)	Total Area (mm²)	N	Conductor Metal	Grounded Conductor Area (mm²)	EGC (Cu)	EGC Area (mm²)	Total Area all Wires (mm²)	Conduit Size (EMT)	Conduit Area (mm²)	Conduit Fill (%)	R _{cu} (Ω/1000')	X _{cu} (Ω/1000')	L (ft)	E (V)	AC Load (A)	Breaker Size (A)	PF (cosθ)	E_VD(APP) (V)	E_VD,LL,APP (%)	E_sLL,APP (V)	E_VD,EX (V)	E_VD,LL,EX (%)	E_sLL,EX (V)	
1	2/0	Al	121.6	364.8	NA	Cu	0	6	38.06	402.9	1-1/2	1314	30.7%	0.1	0.043	150	480	120.3	175	1	3.125	0.65%	476.87	3.13	0.65%	476.87	
2	4	Al	47.1	141.3	NA	Cu	0	8	28.19	169.5	1	556	30.5%	0.3	0.048	250	480	48.1	70	1	6.457	1.35%	473.54	6.46	1.35%	473.54	
3	4	Al	47.1	141.3	NA	Cu	0	8	28.19	169.5	1	556	30.5%	0.3	0.048	50	480	48.1	70	1	1.291	0.27%	478.71	1.29	0.27%	478.71	

MAX. V-DROP FROM PV COMBINER PANEL TO POI

FEEDER No.	Phase Conductors (XHHW-2)					Grounded Conductor/Neutral (XHHW-2)			Equipment Grounding Conductor (XHHW-2)		Conduit Between Inverters and AC Combiner				Parameters for Voltage Drop Calculation								Approximate (APP) EVD = $\sqrt{3}/1000 \times$ (IRLcos θ + IXLsin θ)			Exact (EX) EVD = $\sqrt{3} \times$ [ES + (IRL cos θ + IXL sin θ)/1000 - $\sqrt{[VS^2 - ((IXL \cos \theta - IRL \sin \theta)/1000)^2]}$]		
	L-L-L	No. of Run	Conduit Material	Area (mm ²)	Total Area (mm ²)	N	Conductor Metal	Grounded Conductor Area (mm ²)	EGC (Cu)	EGC Area (mm ²)	Total Area all Wires (mm ²)	Conduit Size (EMT)	Conduit Area (mm ²)	Conduit Fill (%)	R _{cu} (Ω /1000')	X _{cu} (Ω /1000')	L (ft)	E (V)	AC Load (A)	Breaker Size (A)	PF (cos θ)	E _{VD} (APP) (V)	E _{VD,LL,APP} (%)	E _{sLL,APP} (V)	E _{VD,EX} (V)	E _{VD,LL,EX} (%)	E _{sLL,EX} (V)	
C	1/0	2	Al	102.6	307.8	1/0	Cu	117.7	4	52.52	478.0	2	2165	22.1%	0.13	0.044	40	480	216.5	300	1	0.975	0.20%	479.03	0.98	0.20%	479.02	
D	6	1	Al	34.2	102.6	6	Cu	38.06	6	38.06	178.7	1-1/2	1282	13.9%	0.49	0.051	40	4160	24.98	40	1	0.848	0.02%	4159.15	0.85	0.02%	4159.15	

MAX. DC V-DROP FOR HOME RUNS BETWEEN MODULES AND INVERTERS

Inv. No.	Max. Home Run Length (ft.)	PV Wire Guage (AWG)	Wire X-Sectional Area (mm²)	Wire Resistance per 1000' (Ω/kft)	Total Circuit Resistance (Ω)	Modules per String (Mod _{no})	Strings per Inverter (Str _{no})	Module V _{mp} (V)	String Voltage @ V _{mp} (V)	Module I _s (A)	String Current @ I _{mp} (A)	Voltage Drop @ V _{mp} (V)	Voltage Drop @ V _{mp} (%)
1	100	#10	5.26	1.171	0.23	15	16.00	47.89	718.4	13.88	13.15	3.08	0.4%
2	70	#10	5.26	1.171	0.16	15	6.00	47.89	718.4	13.88	13.15	2.16	0.3%
3	200	#10	5.26	1.171	0.47	15	6.00	47.89	718.4	13.88	13.15	6.16	0.9%

MAX. AC + DC V-DROP FOR EACH INVERTER AT POI

Inv. No.	AC V-Drop	FEEDER (C) V-Drop	FEEDER (D) V-Drop	Max. DC V-Drop	Max. Total V-Drop
1	0.65%	0.20%	0.02%	0.40%	1.3%
2	1.35%	0.20%	0.02%	0.30%	1.9%
3	0.27%	0.20%	0.02%	0.90%	1.4%

PV SYSTEM MAX VOLTAGE CALCULATION:

CALCULATION OF Vmax USING TEMPERATURE COEFFICIENT OF Voc of
Q.TRON XL-G2.3 / BFG 630W PV MODULE
TEMPERATURE COEFFICIENT OF Voc = -0.24% / °C
RATED TEMPERATURE = 25°
MINIMUM TEMPERATURE OF SITE AS PER ASHRAE = -27.9°
TEMPERATURE CORRECTION FACTOR = 1+[(−0.0025)x(−27.9-25)] = 1.12696
Vmax = 57.23*1.12696= 64.4959208
Vmax/STRING = 64.4959208*15 = 967.43
MAX INPUT VDC OF SOLIS INVERTER = 1000V

NOTES APPENDIX (AS APPLICABLE FOR TO BE BUILT DRAWING SETS):

- A. TOTAL AC VOLTAGE DROP NOT TO EXCEED 2% TO INTERCONNECTION, < 3% FROM INVERTER(S) TO UTILITY TRANSFORMER.
- B. ALL CONNECTORS 75°C RATED.
- C. ALL CONDUCTORS COPPER, UNLESS OTHERWISE NOTED. DUE TO HIGHER COEFFICIENT OF EXPANSION, ALUMINUM CONDUCTORS REQUIRE MORE MAINTENANCE/INSPECTION THAN COPPER CONDUCTORS. ANNUAL RETORQUEING AS WELL AS INFRARED INSPECTION, MINIMALLY. BE CAREFUL NOT TO CONNECT ALUMINUM WITH COPPER RATED CONDUCTORS OR FITTINGS DURING CONSTRUCTION, TERMINALS SHOULD BE DUAL RATED. MATERIAL FOR NEUTRAL AND EGC SHOULD BE THE SAME IN A GIVEN SEGMENT.
- D. OUTDOOR EQUIPMENT NEMA3R.
- E. ALL CONDUCTORS FOR PV & ESS SYSTEMS MUST BE PROTECTED FROM ACCESS BY A FENCE OR SUITABLE COVER, OR OUT OF REACH.
- F. PROPERTY LINES, BOUNDARIES AND ALL OTHER EXTERIOR MEASUREMENTS ARE FOR REFERENCE ONLY, AND MUST BE VERIFIED BY A LICENSED SURVEYOR OR CIVIL ENGINEER.
- G. ENERGY STORAGE SYSTEMS ARE REQUIRED TO BE INSTALLED IN LOCATIONS AND CONTAINERS IN COMPLIANCE WITH THEIR LISTING REQUIREMENTS.
- H. IF TRAVEL ACROSS A ROOF IS LIMITED TO FIRE SETBACK AREAS, FALL RESTRAINT SYSTEMS MAY BE REQUIRED.
- I. NO PVC ALLOWED ON ROOF OR IN ATTIC.
- J. MC4 CONNECTORS MAY NOT BE JOINED WITH 'MC4 COMPATIBLE' CONNECTORS.
- K. FOR COMMERCIAL SYSTEMS - UNDER MODULE WIRE MANAGEMENT SYSTEMS ARE REQUIRED, RACEWAY FILL MUST NOT EXCEED 40% REFER TO LOCAL REGULATIONS FOR EXCEPTIONS.
- L. TAP CONNECTION IN PANEL MUST NOT VIOLATE CONDITIONS OF ACCEPTABILITY FROM PANEL MANUFACTURER'S NRTL LISTING, OR FIELD LABEL REQUIRED.
- M. PV WIRES MAY NOT BE LAID DIRECTLY ON ROOF, WIRE MANAGEMENT SUCH AS SNAKE TRAY, ETC. MUST BE USED 40% FILL MAX.
- N. TY WRAPS FOR WIRE MANAGEMENT MUST BE STRUCTURAL (S21) UL APPROVED, OR SUN BUNDLER OR EQUAL.
- O. DRAWINGS ARE DIAGRAMMATIC AND INDICATE GENERAL ARRANGEMENT OF SYSTEMS AND WORK. CONDUIT ROUTING, WHEN INDICATED, IS SHOWN DIAGRAMMATICALLY AND DOES NOT SHOW ALL OFFSETS, DROPS, AND RISES OF RUNS. THE CONTRACTOR SHALL ALLOW IN HIS PRICE FOR ROUTING OF CONDUIT TO AVOID OBSTRUCTIONS.
- P. BURIED CONDUITS UNDER AREAS SUBJECT TO VEHICLE TRAFFIC REQUIRE MIN 24" COVER.
- Q. SOLAREEDGE INVERTERS MAY BE EQUIPPED WITH OCPD ON DC LINES, IF NOT EXTERNAL OCPD MAY BE REQUIRED FOR STRINGS NUMBERING 3 OR MORE.
- R. BATTERIES MUST BE IN AN APPROVED BATTERY ENCLOSURE (SPECIFIED BY THE BATTERY MANUFACTURER), SUITABLE FOR THE LOCATION.
- S. NM-B OR PAPER INSULATED CONDUCTORS MAY NOT BE USED EXTERIOR. INSTEAD USE THWN-2 OR EQUAL IN EMT CONDUIT.
- T. FOR MULTIPLE BATTERIES/INVERTERS SHARING A COMMON DC BUS, OCPD PROTECTION IS REQUIRED ON BOTH SIDES OF THE BUS AND AT INVERTERS DUE TO 2 WAY CURRENTS.
- U. THE DEVELOPER IS REQUIRED TO CONFIRM EXISTING ELECTRICAL SERVICE SIZE FROM THE UTILITY, AND MAY NOT RELY SOLELY ON EXISTING BREAKER SIZES.
- V. CONNECTING TO UTILITY EQUIPMENT REQUIRES PRIOR UTILITY CONSENT.
- W. ALL CONNECTIONS MADE FOR FLOATING SOLAR INSTALLATIONS, ABOVE A BODY OF WATER, MUST BE SUBMERSIBLE COMPLIANT.
- X. WHEN BUILDING PV SYSTEMS WITH POWER LINE COMMUNICATIONS FOR RSD SOLUTIONS, FOLLOWING MANUFACTURERS INSTRUCTIONS ON CONDUCTOR AND CONDUIT SPACING IS PARAMOUNT, OR HAZARD MAY RESULT.
- Y. NOTIFY ELECTRICAL ENGINEER WHO'S SEAL IS ON THIS DRAWING PRIOR TO ANY AND ALL CHANGES IN DESIGN.
- Z. PROJECT RELATED SIGNAL/COMMUNICATION CONDUCTORS MAY ONLY BE RUN IN THE SAME CONDUIT WITH POWER CONDUCTORS IF THEIR RATING FOR THE OUTER MOST JACKET IS EQUAL TO OR GREATER THAN THE INSULATION REQUIRED FOR THE HIGHEST VOLTAGE POWER CONDUCTOR CIRCUIT THAT THEY ARE SHARING THE RESPECTIVE CONDUIT WITH.

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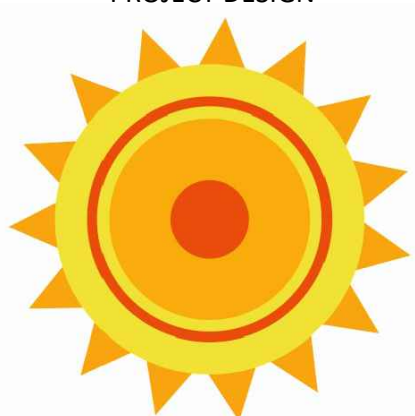
Compact Wire Physical Characteristics (NEC Ch 9 Tbls 5 and 5A)				
AWG	XHHW-2 Al/Cu (Compact Conductors)		XHHW-2 Al/Cu	
	OD (mm)	Area (mm²)	OD (mm²)	Area (mm²)
12	3.669	10.57	3.861	11.68
10	4.247	14.16	4.470	15.68
8	5.690	25.42	5.994	28.19
6	6.604	34.19	6.960	38.06
4	7.747	47.10	8.179	52.52
3	8.394	55.34	8.890	62.06
2	9.144	65.61	9.703	73.94
1	10.540	87.23	11.230	98.97
1/0	11.430	102.60	12.240	117.70
2/0	12.450	121.60	13.410	141.30
3/0	13.720	147.70	14.730	170.50
4/0	14.990	176.30	16.210	206.30
250	16.760	220.70	17.910	251.90
300	18.160	259.00	19.300	292.60
350	19.300	292.60	20.600	333.30
400	20.320	324.30	21.790	373.00
500	22.350	392.40	23.950	450.60
600	24.890	486.60	26.750	561.90
750	27.690	602.00	29.040	679.50
1000	31.240	766.60	33.320	872.20

Al Cond Resistance and Reactance NEC Ch 9 Tbl 9)			
Size	R _{al} (Ω/1000')	R _{cu} (Ω/1000')	X _{al} (Ω/1000')
12	3.200	2.000	0.054
10	2.000	1.200	0.050
8	1.300	0.780	0.052
6	0.810	0.490	0.051
4	0.510	0.310	0.048
3	0.400	0.250	0.047
2	0.320	0.200	0.045
1	0.250	0.160	0.046
1/0	0.200	0.130	0.044
2/0	0.160	0.100	0.043
3/0	0.130	0.082	0.042
4/0	0.100	0.067	0.041
250	0.086	0.057	0.041
300	0.072	0.049	0.041
350	0.062	0.043	0.040
400	0.055	0.038	0.040
500	0.044	0.032	0.039
600	0.037	0.028	0.039
750	0.030	0.024	0.038
1000	0.024	0.019	0.037

PVC Sch 40	Conduit Cross-Sectional Area (NEC Ch 9 Table 4)					
	PVC Sch 80		PVC Sch 40		EMT	
Trade Size	Nom ID	Area	Nom ID	Area	Nom ID	Area
1/2	13.4	141	15.3	184	15.8	196
3/4	18.3	263	20.4	327	20.9	343
1	23.8	445	26.1	535	26.6	556
1-1/4	31.9	799	34.5	935	35.1	968
1-1/2	37.5	1104	40.4	1282	40.9	1314
2	48.6	1855	52.0	2124	52.5	2165
2-1/2	58.2	2660	62.1	3029	69.4	3783
3	72.7	4151	77.3	4693	85.2	5701
3-1/2	84.5	5608	89.4	6277	97.4	7451
4	96.2	7268	101.5	8091	110.1	9521
5	121.1	11518	127.4	12748		
6	145.0	16513	153.2	18433		

Voltage Drop Equations	
Approximate Method	EVD = IRL cosθ + IXL sinθ
Exact Method	EVD = Es + L(IR cosθ + IX sinθ - √Es²2 - (IX cosθ + IR sinθ)²)
Where:	Evd - Measured Voltage Es - Source Voltage I - Current

PROJECT DESIGN

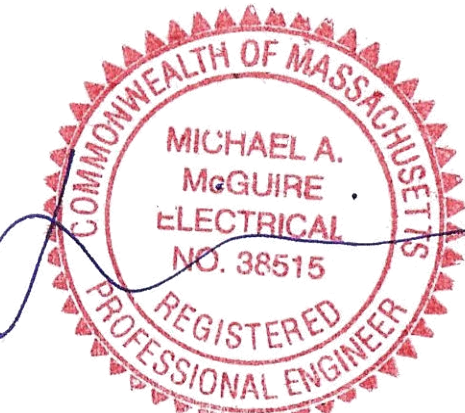


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SEAL/SIGNATURE



sealed 03aug2026 mike@h2dc.com
H2DC PLLC MA CoA#: 001239603
ELECTRICAL ONLY
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PROJECT

BTM ROOFTOP PHOTOVOLTAC SYSTEM

DWARKA LLC

86 SANDERSON AVENUE, LYNN, MA 01902

PROJECT OWNER

DWARKA LLC

PROJECT DEVELOPER



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REVISIONS		
DATE	COMMENT	REV.
5/11/2026	INTERCONNECTION PLAN SET	1.0
8/3/2026	PERMIT PLAN SET	1.0

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PERMIT PLAN SET

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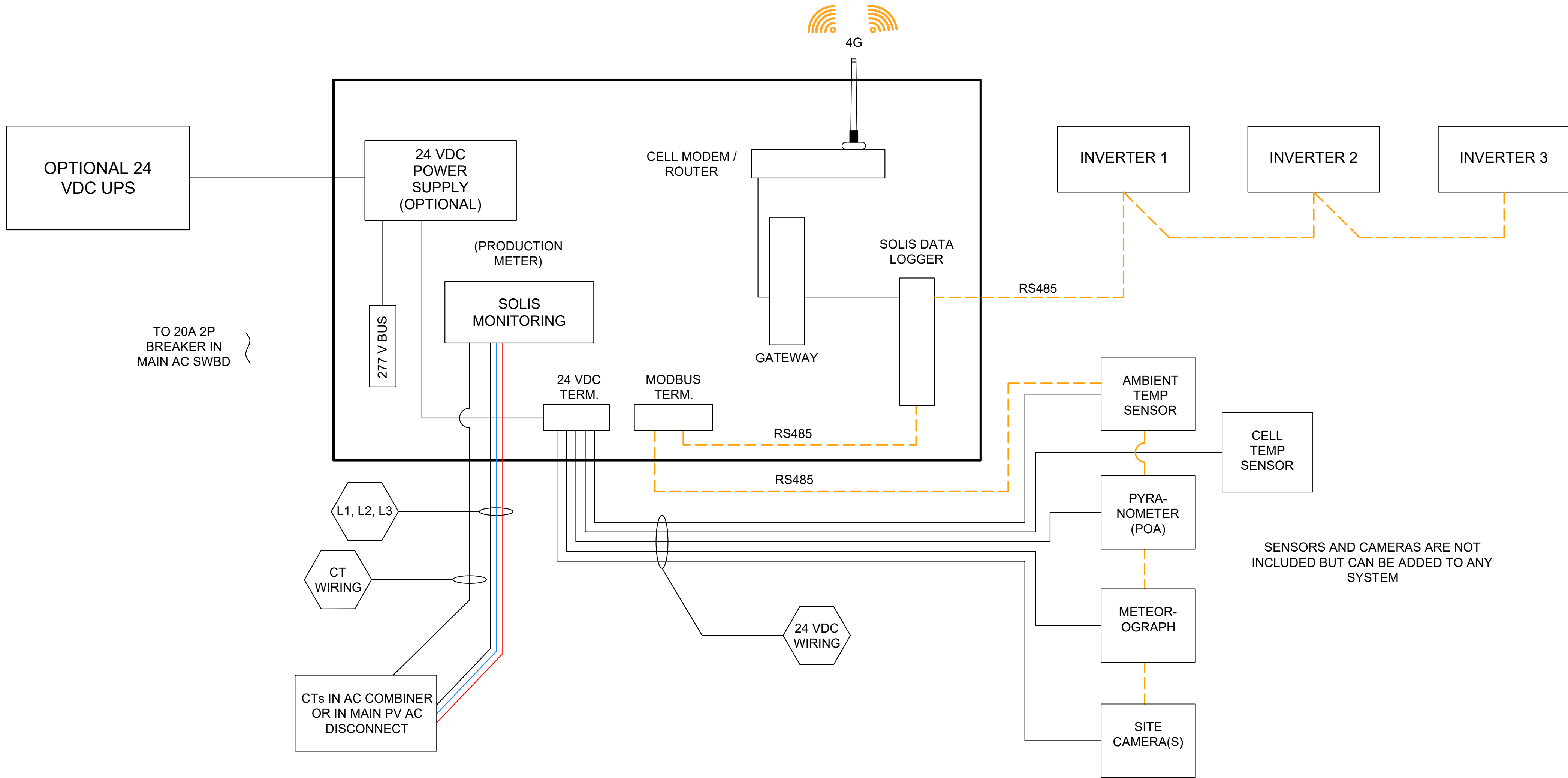
WIRING DETAILS AND CALCULATION

SHEET NUMBER

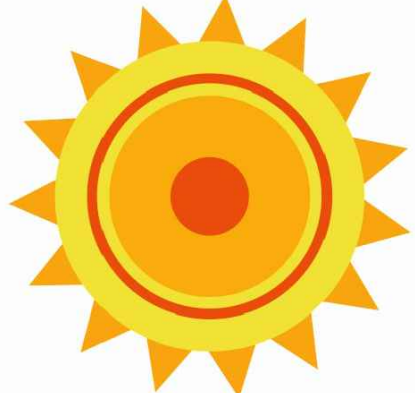
E-102

NOTES APPENDIX (AS APPLICABLE FOR TO BE BUILT DRAWING SETS): (A) TOTAL AC VOLTAGE DROP NOT TO EXCEED 2% TO INTERCONNECTION, < 3% FROM INVERTER(S) TO UTILITY TRANSFORMER. (B) ALL CONNECTORS 75C RATED. (C) ALL CONDUCTORS COPPER, UNLESS OTHERWISE NOTED. DUE TO HIGHER COEFFICIENT OF EXPANSION, ALUMINUM CONDUCTORS REQUIRE MORE MAINTENANCE/INSPECTION THAN COPPER CONDUCTORS. ANNUAL RETORQUEING AS WELL AS INFRARED INSPECTION, MINIMALLY. BE CAREFUL NOT TO CONNECT ALUMINUM WITH COPPER RATED CONDUCTORS OR FITTINGS DURING CONSTRUCTION, TERMINALS SHOULD BE DUAL RATED. MATERIAL FOR NEUTRAL AND EGC SHOULD BE THE SAME IN A GIVEN SEGMENT. (D) OUTDOOR EQUIPMENT NEMA3R. (E) ALL CONDUCTORS FOR PV & ESS SYSTEMS MUST BE PROTECTED FROM ACCESS BY A FENCE OR SUITABLE COVER, OR OUT OF REACH. (F) PROPERTY LINES, BOUNDARIES AND ALL OTHER EXTERIOR MEASUREMENTS ARE FOR REFERENCE ONLY, AND MUST BE VERIFIED BY A LICENSED SURVEYOR OR CIVIL ENGINEER. (G) ENERGY STORAGE SYSTEMS ARE REQUIRED TO BE INSTALLED IN LOCATIONS AND CONTAINERS IN COMPLIANCE WITH THEIR LISTING REQUIREMENTS. (H) IF TRAVEL ACROSS A ROOF IS LIMITED TO FIRE SETBACK AREAS, FALL RESTRAINT SYSTEMS MAY BE REQUIRED. (I) NO PVC ALLOWED ON ROOF OR IN ATTIC. (J) ALL DC CONNECTORS SHALL BE LISTED FOR THE PV APPLICATION. MATING CONNECTORS AT EACH CONNECTION POINT SHALL BE OF THE IDENTICAL MANUFACTURER AND MODEL. MC4-COMPATIBLE CONNECTORS FROM DIFFERENT MANUFACTURERS SHALL NOT BE MATED WITH EACH OTHER. 'MC4-COMPATIBLE' AND 'ORIGINAL MC4' SHALL NOT BE MATED WITH EACH OTHER. ANY EXCEPTION REQUIRES WRITTEN COMPATIBILITY DOCUMENTATION FROM BOTH MANUFACTURERS PRIOR TO INSTALLATION. (K) FOR COMMERCIAL SYSTEMS - UNDER MODULE WIRE MANAGEMENT SYSTEMS ARE REQUIRED, RACEWAY FILL MUST NOT EXCEED 40% REFER TO LOCAL REGULATIONS FOR EXCEPTIONS. (L) TAP CONNECTION IN PANEL MUST NOT VIOLATE CONDITIONS OF ACCEPTABILITY FROM PANEL MANUFACTURER'S NRTL LISTING, OR FIELD LABEL REQUIRED. (M) PV WIRES MAY NOT BE LAID DIRECTLY ON ROOF, WIRE MANAGEMENT SUCH AS SNAKE TRAY, ETC. MUST BE USED 40% FILL MAX. (N) TY WRAPS FOR WIRE MANAGEMENT MUST BE STRUCTURAL (S21) UL APPROVED, OR SUN BUNDLER OR EQUAL. (O) DRAWINGS ARE DIAGRAMMATIC AND INDICATE GENERAL ARRANGEMENT OF SYSTEMS AND WORK. CONDUIT ROUTING, WHEN INDICATED, IS SHOWN DIAGRAMMATICALLY AND DOES NOT SHOW ALL OFFSETS, DROPS, AND RISES OF RUNS. THE CONTRACTOR SHALL ALLOW IN HIS PRICE FOR ROUTING OF CONDUIT TO AVOID OBSTRUCTIONS. (P) BURIED CONDUITS UNDER AREAS SUBJECT TO VEHICLE TRAFFIC REQUIRE MIN 24" COVER. (Q) SOLAREEDGE INVERTERS MAY BE EQUIPPED WITH OCPD ON DC LINES, IF NOT EXTERNAL OCPD MAY BE REQUIRED FOR STRINGS NUMBERING 3 OR MORE. (R) BATTERIES MUST BE IN AN APPROVED BATTERY ENCLOSURE (SPECIFIED BY THE BATTERY MANUFACTURER), SUITABLE FOR THE LOCATION. (S) NM-B OR PAPER INSULATED CONDUCTORS MAY NOT BE USED EXTERIOR. INSTEAD USE THWN-2 OR EQUAL IN EMT CONDUIT. (T) FOR MULTIPLE BATTERIES/INVERTERS SHARING A COMMON DC BUS, OCPD PROTECTION IS REQUIRED ON BOTH SIDES OF THE BUS AND AT INVERTERS DUE TO 2 WAY CURRENTS. (U) THE DEVELOPER IS REQUIRED TO CONFIRM EXISTING ELECTRICAL SERVICE SIZE FROM THE UTILITY, AND MAY NOT RELY SOLELY ON EXISTING BREAKER SIZES. (V) CONNECTING TO UTILITY EQUIPMENT REQUIRES PRIOR UTILITY CONSENT. (W) ALL CONNECTIONS MADE FOR FLOATING SOLAR INSTALLATIONS, ABOVE A BODY OF WATER, MUST BE SUBMERSIBLE COMPLIANT. (X) WHEN BUILDING PV SYSTEMS WITH POWER LINE COMMUNICATIONS FOR RSD SOLUTIONS, FOLLOWING MANUFACTURERS INSTRUCTIONS ON CONDUCTOR AND CONDUIT SPACING IS PARAMOUNT, OR HAZARD MAY RESULT. (Y) NOTIFY ELECTRICAL ENGINEER WHO'S SEAL IS ON THIS DRAWING PRIOR TO ANY AND ALL CHANGES IN DESIGN. (Z) PROJECT RELATED SIGNAL/COMMUNICATION CONDUCTORS MAY ONLY BE RUN IN THE SAME CONDUIT WITH POWER CONDUCTORS IF THEIR RATING FOR THE OUTER MOST JACKET IS EQUAL TO OR GREATER THAN THE INSULATION REQUIRED FOR THE HIGHEST VOLTAGE POWER CONDUCTOR CIRCUIT THAT THEY ARE SHARING THE RESPECTIVE CONDUIT WITH. (AA) EQUIPMENT MUST BE LOCATED AT LEAST 1 FOOT ABOVE 100-YEAR FLOOD LEVEL.

DAS ENCLOSURE TO BE MOUNTED ON EXTERIOR WALL ADJACENT TO PV DISCONNECT



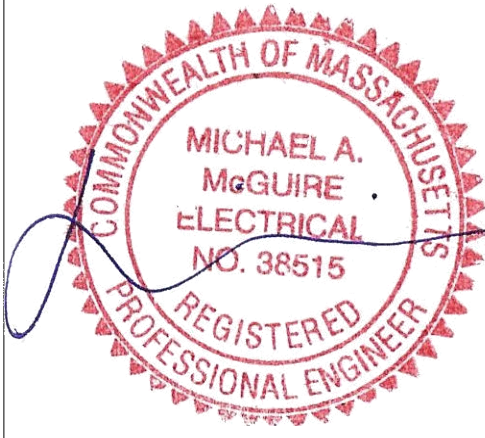
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sealed 03aug2026 mike@h2dc.com
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PROJECT

BTM ROOFTOP PHOTOVOLTAIC SYSTEM
DWARKA LLC
86 SANDERSON AVENUE, LYNN, MA 01902

PROJECT OWNER
DWARKA LLC

PROJECT DEVELOPER



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PERMIT PLAN SET

SHEET TITLE
DATA ACQUISITION SYSTEM

SHEET NUMBER

E-103

INVERTER 1
PHOTOVOLTAIC SYSTEM
DC DISCONNECT

WARNING

ELECTRIC SHOCK HAZARD
DO NOT TOUCH TERMINALS
TERMINALS ON BOTH THE LINE AND LOAD SIDES
MAY BE ENERGIZED IN THE OPEN POSITION

INTERACTIVE SOLAR PV SYSTEM RATING

RATED DC CURRENT	210.40 A
RATED DC VOLTAGE	858.45 VDC
MAXIMUM SYSTEM VOLTAGE	967.43 VDC
SHORT CIRCUIT CURRENT	222.08 A

4" X 4" RED/WHITE PLACARD (QTY. 1)
LABEL LOCATION: APPLY TO OR NEAR DC
DISCONNECT AT INVERTERS #1.

INVERTER 2
PHOTOVOLTAIC SYSTEM
DC DISCONNECT

WARNING

ELECTRIC SHOCK HAZARD
DO NOT TOUCH TERMINALS
TERMINALS ON BOTH THE LINE AND LOAD SIDES
MAY BE ENERGIZED IN THE OPEN POSITION

INTERACTIVE SOLAR PV SYSTEM RATING

RATED DC CURRENT	78.90 A
RATED DC VOLTAGE	858.45 VDC
MAXIMUM SYSTEM VOLTAGE	967.43 VDC
SHORT CIRCUIT CURRENT	83.28 A

4" X 4" RED/WHITE PLACARD (QTY. 1)
LABEL LOCATION: APPLY TO OR NEAR DC
DISCONNECT AT INVERTERS #2.

INVERTER 3
PHOTOVOLTAIC SYSTEM
DC DISCONNECT

WARNING

ELECTRIC SHOCK HAZARD
DO NOT TOUCH TERMINALS
TERMINALS ON BOTH THE LINE AND LOAD SIDES
MAY BE ENERGIZED IN THE OPEN POSITION

INTERACTIVE SOLAR PV SYSTEM RATING

RATED DC CURRENT	78.90 A
RATED DC VOLTAGE	858.45 VDC
MAXIMUM SYSTEM VOLTAGE	967.43 VDC
SHORT CIRCUIT CURRENT	83.28 A

4" X 4" RED/WHITE PLACARD (QTY. 1)
LABEL LOCATION: APPLY TO OR NEAR DC
DISCONNECT AT INVERTERS #3.

SOLAR PV SYSTEM
EQUIPPED WITH RAPID
SHUTDOWN

TURN THE RAPID SHUTDOWN
SWITCH TO THE "OFF"
POSITION TO SHUT DOWN
CONDUCTORS LEAVING THE
ARRAY. DC CONDUCTORS
WITHIN THE ARRAY REMAIN
ENERGIZED IN SUNLIGHT. THE
INVERTER WILL LIMIT VOLTAGE
LEAVING THE ARRAY TO 30V
WITHIN 30 SECONDS OF RAPID
SHUTDOWN INITIATION.

SOLAR PV SYSTEM
EQUIPPED WITH RAPID
SHUTDOWN

TURN THE RAPID SHUTDOWN
SWITCH TO THE "OFF"
POSITION TO SHUT DOWN
CONDUCTORS LEAVING
THE ARRAY. DC CONDUCTORS
WITHIN THE ARRAY REMAIN
ENERGIZED IN SUNLIGHT. THE
INVERTER WILL LIMIT VOLTAGE
LEAVING THE ARRAY TO 30V WITHIN
30 SECONDS OF RAPID SHUTDOWN
INITIATION. THE MLPEs WILL LIMIT
THE VOLTAGE OF THE SUB-ARRAY
TO 80V WITHIN 30 SECONDS

3" X 6" VINYL LABEL (QTY. 4)
LABEL LOCATION: PV INVERTER 1, 2 & 3, PV
DISCONNECT SWITCH
(PER CODE: CEC690.56(C)(1))

**CAUTION: SOLAR ELECTRIC
SYSTEM CONNECTED**

6" X 1 1/4" BLACK ON YELLOW PLACARD (QTY. 5)
LABEL LOCATION: APPLY AT POI, ON UTILITY METER CABINET,
SWITCHBOARD, TRANSFORMER CABINET
(PER CODE: CEC690.15, 690.13(B)).

DANGER

**DUAL POWER
SOURCE**

**SECOND SOURCE
IS PHOTOVOLTAIC
SYSTEM**

2 3/4" X1 5/8" VINYL LABEL (QTY. 10)
LABEL LOCATION: TRANSFORMER, POI,
METER CABINET, SWITCHBOARD
(PER CODE: CEC 705.12(D)(4),
NEC 705.30(C) & NEC 690.59)

WARNING

PHOTOVOLTAIC POWER SOURCE

5 3/4" X 1 1/8" VINYL LABEL (QTY. 30)
LABEL LOCATION: CONDUIT, AC COMBINER, SERVICE EQUIPMENT
(PER CODE: NEC690.31(G)(3)(4) & NEC 690.13(G)(4))

WARNING

ARC FLASH AND SHOCK HAZARD
Appropriate PPE Required

Do not operate controls or open covers
without appropriate personel protection
equipment.

Failure to comply may result in injury or
death!

Refer to NFPA 70E for minimum PPE requirements

5" X 3 1/2" VINYL LABEL (QTY. 20)
LABEL LOCATION: PV AC COMBINER, PV AC DISCONNECT
SWITCHBOARD, ANY/ALL SERVICE EQUIPMENT
(PER CODE: NEC706.15(C) and NEC 110.16 and
NFPA 70E Article 130.5(C)(1),(2),(3))

WARNING

ELECTRIC SHOCK HAZARD
DO NOT TOUCH TERMINALS
TERMINALS ON BOTH LINE & LOAD SIDES
MAY BE ENERGIZED IN OPEN POSITION
DO NOT DISCONNECT FUSES UNDER LOAD

THE DC CONDUCTORS OF THIS
PHOTOVOLTAIC SYSTEM ARE
UNGROUNDDED AND MAY BE ENERGIZED

PHOTOVOLTAIC SYSTEM
DC DISCONNECT

AUTHORIZED PERSONNEL ONLY

Note: WARNING labels must resemble format
in example above with over-sized WARNING,
exclamation point in triangle, colors, etc.

CAUTION

MULTIPLE SOURCES OF POWER

↑
N

PV AC DISCONNECT
UTILITY BREAKER →

IN CASE OF
LIFE-THREATENING
EMERGENCY
PLEASE DIAL 911

EMERGENCY PV
SYSTEM CONTACT: (617)
842-8016 SURYA
STANDARD ENERGY LLC

5" X 4" VINYL LABEL (QTY. 6)
LABEL LOCATION: APPLY TO PV AC DISCONNECT, ADJACENT TO
EPO SWITCHES, SWITCHBOARD NEAR MAIN BREAKER, UTILITY
TRANSFORMER, AND ON BOTH THE INTERIOR AND EXTERIOR
ELECTRICAL ROOM DOORS. (PER CODE: NEC 705.10 & NEC
710.10)

WARNING

ELECTRIC SHOCK HAZARD
DO NOT TOUCH TERMINALS
TERMINALS ON BOTH THE LINE AND
LOAD SIDES MAY BE ENERGIZED
IN THE OPEN POSITION

PHOTOVOLTAIC MODULES PRODUCE DC VOLTAGE
WHENEVER THEY ARE EXPOSED TO SUNLIGHT

4" X 3" VINYL LABEL (QTY. 20)
LABEL LOCATION: PV AC DISCONNECT, INDIVIDUAL
INVERTER AC DISCONNECTS, PV AC
COMBINER, SWITCHBOARD, TRANSFORMER,
POINT OF INTERCONNECTION.
(PER CODE: NEC 690.17(E)), (CB))

WARNING

ELECTRIC SHOCK HAZARD
TERMINALS ON THE LINE AND
LOAD SIDES MAY BE ENERGIZED
IN THE OPEN POSITION

4" X 2" VINYL LABEL (QTY. 20)
LABEL LOCATION: PV AC DISCONNECT, POINT
OF INTERCONNECTION - ALTERNATIVE FOR
LABEL ABOVE
(PER CODE: NEC 690.17(E)), (CB))

WARNING

ELECTRIC SHOCK HAZARD
THE DC CONDUCTORS OF THIS
PHOTOVOLTAIC SYSTEM
ARE UNGROUNDDED AND MAY
BE ENERGIZED

2 3/4" X1 5/8" VINYL LABEL (QTY. 30)
LABEL LOCATION: INDIVIDUAL
INVERTER DC DISCONNECTS.
(PER CODE: NEC 690.35(F))

CAUTION SOLAR CIRCUIT

6" X 1 1/8" VINYL LABEL (QTY. 100)
LABEL LOCATION: MARKINGS PLACED ON ALL INTERIOR AND EXTERIOR
DC CONDUIT, RACEWAYS, ENCLOSURES, AND CABLE ASSEMBLIES AT LEAST
EVERY 10 FT, AT TURNS AND ABOVE/BELOW PENETRATIONS AND ALL
COMBINER/JUNCTION BOXES.
(PER CODE: IFC605.11.1.4)

WARNING

DUAL POWER SUPPLY
SOURCES: UTILITY GRID AND
PV SOLAR ELECTRIC SYSTEM

TURN OFF PHOTOVOLTAIC AC DISCONNECT
PRIOR TO WORKING INSIDE PANEL

2 3/4" X1 5/8" VINYL LABEL (QTY. 30)
LABEL LOCATION: ALL EXISTING
DISTRIBUTION PANELS
(PER CODE: NEC 705.30(C) &
NEC 690.59); NEC 110.27(C) &
OSHA 1910.145(f)(7)

WARNING

**TURN OFF PHOTOVOLTAIC
AC DISCONNECT PRIOR TO
WORKING INSIDE PANEL**

4" X 2" VINYL LABEL (QTY. 20)
LABEL LOCATION: ALL EXISTING
DISTRIBUTION PANELS
(PER CODE: NEC 110.27(C) & OSHA 1910.145(f)(7))

UTILITY SERVICE METER

3" X 1/2" PERMANENT PLAQUE (QTY. 1)
LABEL LOCATION : UTILITY METER

UTILITY RTU CABINET

3" X 1/2" PERMANENT PLAQUE (QTY. 1)
LABEL LOCATION : UTILITY METER

UTILITY PV GENERATION

3" X 1/2" PERMANENT PLAQUE (QTY. 1)
LABEL LOCATION : PV METER

PV TRANSFORMER

3" X 1/2" PERMANENT PLAQUE (QTY. 1)
LABEL LOCATION : PV XFMR

UTILITY TRANSFORMER

3" X 1/2" PERMANENT PLAQUE (QTY. 1)
LABEL LOCATION : UTILITY XFMR

INVERTER 1

INVERTER 2

INVERTER 3

3" X 1" BLACK PLACARD WITH
WHITE LETTERING (QTY. 3) LABEL
LOCATION: APPLY TO INVERTERS
#1, 2 AND 3

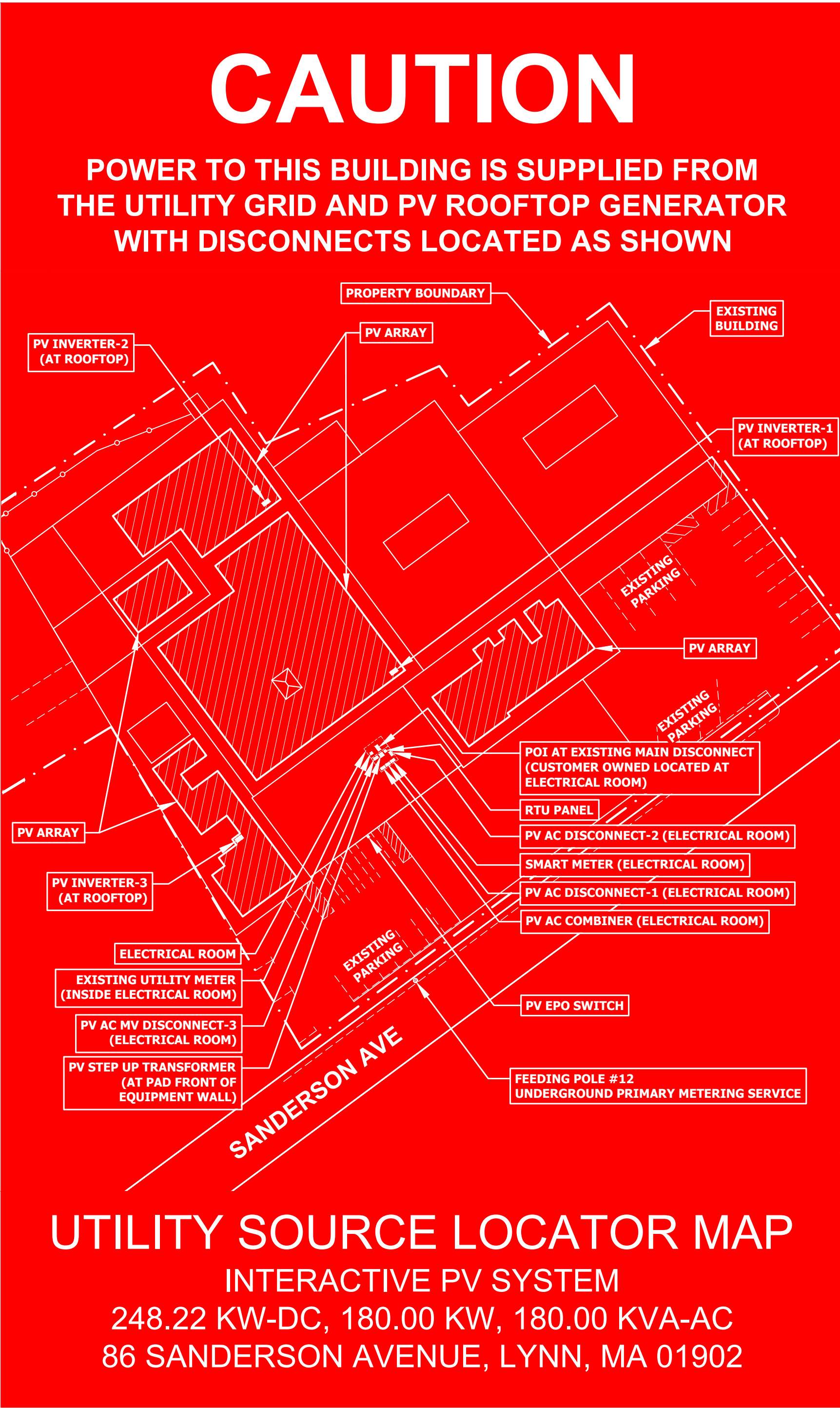
**PHOTOVOLTAIC SYSTEM
MAIN AC DISCONNECT**

AC OPERATING CURRENT: 24.98 A
NOMINAL AC OPERATING VOLTAGE: 4.16 kV

WARNING

ELECTRIC SHOCK HAZARD
AUTHORIZED PERSONNEL ONLY DO
NOT TOUCH TERMINALS, BOTH LINE
AND LOAD SIDE MAY BE ENERGIZED
IN THE OPEN POSITION

5" X 5" RED/WHITE PLACARD (QTY. 1)
LABEL LOCATION: APPLY TO MAIN PV DISCONNECT.



UTILITY SOURCE LOCATOR MAP
INTERACTIVE PV SYSTEM
248.22 KW-DC, 180.00 KW, 180.00 KVA-AC
86 SANDERSON AVENUE, LYNN, MA 01902

6" X 10" OR 9" X 15" WHITE ON RED ENGRAVED PLACARD (QTY. 3)
LABEL LOCATION: MAIN PV AC DISCONNECT, AC COMBINER, MAIN SWITCHBOARD
(PER CODE: NEC 705.10 & NEC 710.10)

PROJECT DESIGN

SURYA
Standard
Harnessing the best source of energy available

3 Allied Drive, Suite 107
Dedham, MA 02026
amanullah@suryastandard.com
(617) 842-8016

SEAL/SIGNATURE

COMMONWEALTH OF MASSACHUSETTS

MICHAEL A. McGUIRE
ELECTRICAL
NO. 38615

REGISTERED
PROFESSIONAL ENGINEER

sealed 03aug2026 mike@h2dc.com
H2DC PLLC MA CoA#: 001239603
ELECTRICAL ONLY
FOR PERMITTING PURPOSES ONLY

PROJECT
**BTM ROOFTOP
PHOTOVOLTAIC SYSTEM**
DWARKA LLC
86 SANDERSON AVENUE, LYNN, MA 01902

PROJECT OWNER
DWARKA LLC

PROJECT DEVELOPER

SURYA
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Harnessing the best source of energy available

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Dedham, MA 02026
amanullah@suryastandard.com
(617) 842-8016

REVISIONS		
DATE	COMMENT	REV.
5/11/2026	INTERCONNECTION PLAN SET	1.0
8/3/2026	PERMIT PLAN SET	1.0

PROJECT NO:
FILE NAME: DWARKA LLC
DRAWN BY: SPV
CHECKED BY:
COPYRIGHT:

PERMIT PLAN SET

SHEET TITLE
**PLACARD AND
LABEL**

SHEET NUMBER
E-201

FOR PERMITTING PURPOSES ONLY



AC Combiner Panels - 400 to 1000A, 480-600VAC



Custom Panels UL Approved for PV Applications

UL certified AC combiner panels custom designed and manufactured for string inverter applications. UL 891, design and construction, UL 508A certification. These panels are thermally engineered for demanding solar applications, utilizing electronic trip MCCBs and painted our standard gloss white color.

Wall Mount 480VAC to 600VAC, 400-800A, MLO Output AC Combiner

- Combines two to eight string inverters
- 480VAC – 600VAC 3PH, 3W or 4W
- 400A to 1,000A bus rating, MLO
- Up to 250A Main MCCB option
- 15A to 250A MCCBs, TM or ETU trip, 80% or 100% rated
- Interrupt ratings up to 50kAIC @ 600VAC, 50kAIC @ 480VAC
- Incoming cable size up to 350MCM
- Outgoing cable up to 750MCM (3) cables per phase
- NEMA 3R wall mount and NEMA 4 roof mount
- UL 891 const / UL 508A certification (N3R)
- UL508A const / UL 508A certification (N4)



Pictured above: UL / NEMA 3R, 800A AC combiner behind N3R door, dead front view



Pictured above: UL / NEMA 3R, 800A AC combiner behind N3R door, behind dead front view.



Pictured above: UL / NEMA 4, 800A AC combiner mounted on rooftop.

Electrical Panels - Engineering - Controls
Visit our site: www.epecsolutionsinc.com

by EPEC Solutions, Inc.
Contact: info@epecsolutionsinc.com



AC Combiner Panels – 400 To 1000A, 480-600VAC

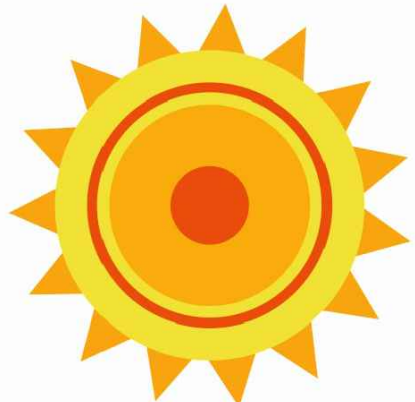
Custom Panels UL Approved for PV Applications

UL certified AC combiner panels custom designed and manufactured for string inverter applications. UL 891, design and construction, UL 508A certification. These panels are thermally engineered for demanding solar applications, utilizing electronic trip MCCBs and painted our standard gloss white color.

Wall Mount 480VAC to 600VAC, 400-800A, MLO Output AC Combiner

- Combines two to eight string inverters
- 480VAC — 600VAC 3PH, 3W or 4W
- 400A to 1,000A bus rating, MLO
- Up to 250A Main MCCB option
- 15A to 250A MCCBs, TM or ETU trip, 80% or 100% rated
- Interrupt ratings up to 50kAIC @ 600VAC, 50kAIC @ 480VAC
- Incoming cable size up to 350MCM
- Outgoing cable up to 750MCM (3) cables per phase
- NEMA 3R wall mount and NEMA 4 roof mount
- UL 891 const / UL 508A certification (N3R)
- UL508A const / UL 508A certification (N4)

PROJECT DESIGN



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3 Allied Drive, Suite 107
Dedham, MA 02026
amanullah@suryastandard.com
(617) 842-8016

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PROJECT

**BTM ROOFTOP
PHOTOVOLTAC SYSTEM**

DWARKA LLC

86 SANDERSON AVENUE, LYNN, MA 01902

PROJECT OWNER

DWARKA LLC

PROJECT DEVELOPER



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PROJECT NO:

FILE NAME: DWARKA LLC

DRAWN BY: SPV

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PERMIT PLAN SET

SHEET TITLE

**PV COMBINER PANEL
SPECIFICATION**

SHEET NUMBER

E-204



Quote Q236571-2
6/11/2026

Contact Your Sales Rep
Ben Meade
Phone: (360) 667-2285
ben.meade@maddox.com

Quote To:
Surya Standard Energy LLC
3 Allied Drive

Pay Terms: To be determined

Ship To:
TBD

#	Description	Count	Price	Amount
1	General Purpose Dry-Type Transformer New 225 kVA 3-Ph Solar Duty Dry-Type Transformer Primary Voltage: 4160 Y 2400 Secondary Voltage: 480 Y 277 Taps: 2A / B @2.5% Frequency: 60 Hz Temperature Rise: 150°C Cooling Class: AA Conductor: Al / Al Enclosure: Outdoor NEMA 3R Shipping: Freight Allowed Warranty: 5 Years Lead Time: Made to Order, 18-20 weeks			



Quote Lines	Additional Cost	Total Quote
Before Tax		
Tax Estimate		
Total		

Notes:
This quote may include products that are in full or in part subject to price adjustment due to U.S. import tariffs or duties applicable to the quoted product.
Buyer is responsible for any increases due to changes in government-imposed fees.

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**BTM ROOFTOP
PHOTOVOLTAIC SYSTEM**

DWARKA LLC
86 SANDERSON AVENUE, LYNN, MA 01902

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FILE NAME: DWARKA LLC

DRAWN BY: SPV

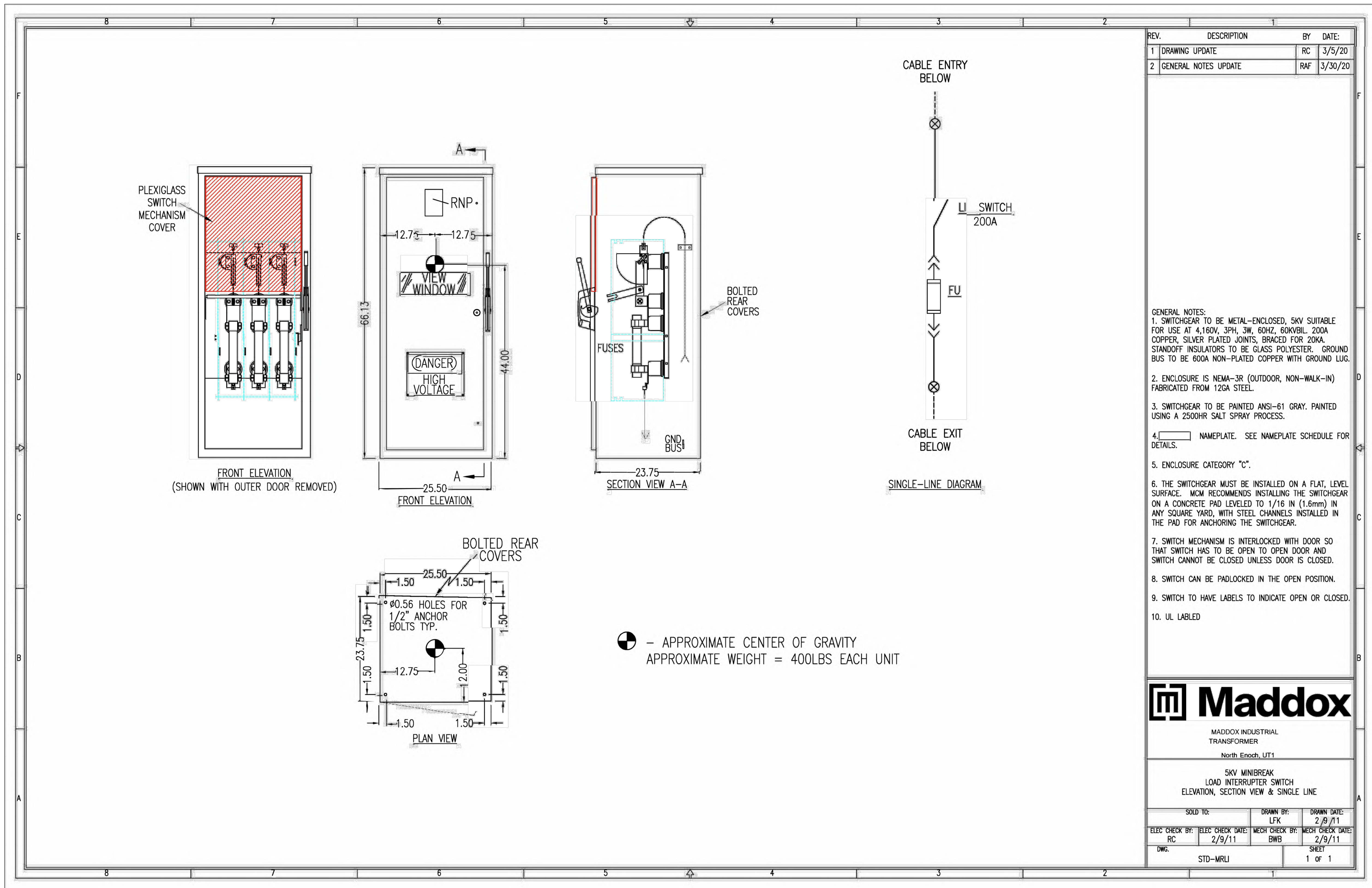
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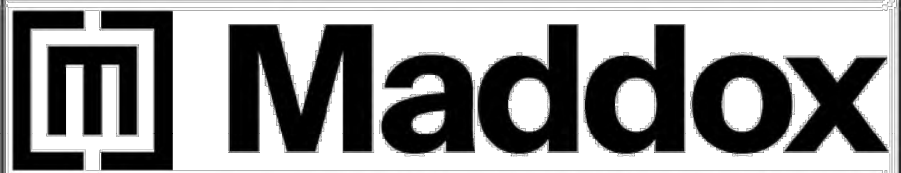
SHEET TITLE
**PV TRANSFORMER
SPECIFICATION**

SHEET NUMBER
E-205



REV.	DESCRIPTION	BY	DATE:
1	DRAWING UPDATE	RC	3/5/20
2	GENERAL NOTES UPDATE	RAF	3/30/20

- GENERAL NOTES:
1. SWITCHGEAR TO BE METAL-ENCLOSED, 5KV SUITABLE FOR USE AT 4.160V, 3PH, 3W, 60HZ, 60KVBL. 200A COPPER, SILVER PLATED JOINTS, BRACED FOR 20KA. STANDOFF INSULATORS TO BE GLASS POLYESTER. GROUND BUS TO BE 600A NON-PLATED COPPER WITH GROUND LUG.
 2. ENCLOSURE IS NEMA-3R (OUTDOOR, NON-WALK-IN) FABRICATED FROM 12GA STEEL.
 3. SWITCHGEAR TO BE PAINTED ANSI-61 GRAY. PAINTED USING A 2500HR SALT SPRAY PROCESS.
 4. NAMEPLATE. SEE NAMEPLATE SCHEDULE FOR DETAILS.
 5. ENCLOSURE CATEGORY "C".
 6. THE SWITCHGEAR MUST BE INSTALLED ON A FLAT, LEVEL SURFACE. MCM RECOMMENDS INSTALLING THE SWITCHGEAR ON A CONCRETE PAD LEVELED TO 1/16 IN (1.6mm) IN ANY SQUARE YARD, WITH STEEL CHANNELS INSTALLED IN THE PAD FOR ANCHORING THE SWITCHGEAR.
 7. SWITCH MECHANISM IS INTERLOCKED WITH DOOR SO THAT SWITCH HAS TO BE OPEN TO OPEN DOOR AND SWITCH CANNOT BE CLOSED UNLESS DOOR IS CLOSED.
 8. SWITCH CAN BE PADLOCKED IN THE OPEN POSITION.
 9. SWITCH TO HAVE LABELS TO INDICATE OPEN OR CLOSED.
 10. UL LABELED



MADDOX INDUSTRIAL TRANSFORMER North Enoch, UT1			
5KV MINIBREAK LOAD INTERRUPTER SWITCH ELEVATION, SECTION VIEW & SINGLE LINE			
SOLD TO:	DRAWN BY: LFK	DRAWN DATE: 2/9/11	
ELEC CHECK BY: RC	ELEC CHECK DATE: 2/9/11	MECH CHECK BY: BWB	MECH CHECK DATE: 2/9/11
DWG. STD-MRLI	SHEET 1 OF 1		

PROJECT DESIGN

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SEAL/SIGNATURE

PROJECT
**BTM ROOFTOP
PHOTOVOLTAIC SYSTEM**
DWARKA LLC
86 SANDERSON AVENUE, LYNN, MA 01902
PROJECT OWNER
DWARKA LLC

PROJECT DEVELOPER

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8/3/2026	PERMIT PLAN SET	1.0

PROJECT NO:
FILE NAME: DWARKA LLC
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PERMIT PLAN SET

SHEET TITLE
**PV MV DISCONNECT
SPECIFICATION**

SHEET NUMBER
E-206

Procedural Notes

1. Prior to commencement of any work, the contractor shall notify owner, owner's representative and Architect/Engineer of any discrepancies noted among site conditions, manufacturer recommendations or codes, regulations or rules of jurisdictions having authority.
2. All dimensions of existing conditions must be verified prior to commencing work.
3. The contractor is responsible for all bracing and shoring of equipment during installation.
4. Contractor shall be responsible for all safety precautions and measures on site. The Architect/Engineer has no overall supervisory authority and no direct responsibility for the specific working conditions.
5. Contractor initiated changes shall be submitted in writing to the Architect/Engineer for approval prior to making any changes. Approved changes shall require a drawing revision to maintain control over the Architect/Engineer approved design. Deviation from these plans prior to Architect/Engineer approval places the contractor at risk.

General Notes

1. The electrical contractor is responsible for installing all equipment and following all directions and instructions shown herein.
2. The electrical contractor is advised that all drawings, component manuals, especially the inverter manuals, are to be read and understood prior to installation. The contractor is also advised to have all component switches in the off position and fuses removed prior to installation of fuse-bearing components and lock-out all disconnects.
3. For safety it is recommended the installation crew always have a minimum of two people working together.
4. This solar photovoltaic system is to be installed following the conventions of the National Electric Code as accepted/amended by the local AHJ. Any local code which may supersede the NEC shall govern.
5. All electrical components to be installed with this system are to be "UL" listed or accepted equivalent. Equipment shall be NEMA 3R outdoor rated or better, unless located indoors.
6. The contractor is responsible for selecting and purchasing equipment that will last the lifetime of the PV system (20 years min.). All enclosures, conduit straps, painted metal surfaces, concrete, grounding equipment and other products shall be selected to last the lifetime of the PV system. The Architect/Engineer specifies the minimum required equipment and specifications to accomplish the project and the electrical contractor is responsible to ensure that these specifications are met or exceeded with good quality equipment, workmanship and skill.
7. DC voltage from the array is always present at the DC disconnect enclosure and the DC terminals of the inverter during daylight hours. All persons working or involved with this photovoltaic system are warned that the solar modules are energized whenever they are exposed to daylight.
8. All portions of this solar electric system shall be clearly marked in accordance with the latest electrical code in effect for the project site.
9. For proper maintenance and isolation of inverters, refer to isolation procedure in inverter operation manual.
10. This photovoltaic system's utility interconnection point shall meet the specific requirements of Articles 690 and 705 of the National Electrical Code and the requirements of the local electrical utility of authority in this jurisdiction.
11. The grounding of the photovoltaic system shall comply with Article 690 of the National Electrical Code.
12. The electrical contractor is not to start or energize the PV or inverter system at any time until approved by the owner and governing electric utility.
13. The contractor is responsible for mounting all equipment per the manufacturer's specifications. If specifications are not apparent, the contractor shall use diligent efforts to mount equipment such that it will be clean, level and solid in order to last the lifetime of this solar electric system.
14. Any metal shavings resulting from site work shall be cleaned from enclosure interiors, top surfaces of enclosures. The ground surface and any additional areas where oxidized or conductive metal shavings may cause rust, electrical short circuits or other damage.
15. The electrical contractor shall consider the weathering of equipment over time and eliminate the possibility of degradation of equipment due to water entry and UV exposure. As a result, we require the use of unistrut or similar mounting systems to mount enclosures, pull boxes, load centers, fuse boxes or other equipment to rooftops and walls to prevent water build-up may occur. Weep holes shall be provided in enclosures where condensation or water build-up may occur. Sealing conduit with a fire retardant foam or caulk at enclosure entry points is recommended to minimize condensation and pests in enclosures.
16. The contractor will provide submittals (including drawings, catalog cuts, and manufacturer's data, etc.) for all equipment and materials. Submittals will be reviewed and returned. The contractor will be at risk for any work related to un-approved submittals.

Electrical Notes

1. In every pull box, terminal box, and at all places where wires may not be readily identified by nameplate markings on the equipment to which they connect, identify each circuit with a plastic label or tag.
2. The layout of conduit shown in these plans is indicative only. Contractor shall route and locate the conduits to suit site conditions. Contractor will submit for approval all changes in wiring and conduit with the Architect/Engineer which exceeds noted length of run.
3. Where wire and cable routing is not shown, and destination only is indicated. Contractor shall determine exact routing and lengths required. A shop drawing of proposed installation shall be supplied to owner and owner's representative prior to installation and include proposed length of run.
4. Underground conduit shall be UV resistant outdoor rated PVC Schedule 40. Conduit types are only recommended on the electrical diagrams. It is the responsibility of the electrical contractor to conform to these requirements.
5. All metallic connectors and fittings shall be non-corroding, such as aluminum, stainless steel or galvanized.
6. Bends shall not damage the raceway or significantly change the internal diameter of raceways (no kinks).
7. Support conductors in vertical conduits in accordance with NEC requirements.
8. Install all wiring materials in a neat workmanlike manner. Use good trade practices as required by chapter 3 of the NEC.
9. Arrange conduit to maintain headroom and in a neat inconspicuous manner. Run parallel and at right angles to structural members. Provide boxes, fittings and bends for change in direction. Fasten conduit securely in place.
10. Support conduit using steel or malleable iron straps, lay-in adjustable hangers, clevis hangers and split-hangers. Hanger spacing shall be 10' maximum. Use approved beam clamps for connection to structural members.
11. Provide pull and junction boxes where required to facilitate the installation of wiring in addition to those shown on the drawings. Bends in conduits between pull boxes shall not exceed the equivalent of four 90 degree bends.
12. When field cutting is required, the conduit shall be cut square and deburred.
13. Conduit sizes not specified should conform to NEC specifications. Minimum conduit size 1" unless noted otherwise.
14. The wiring minimum size should be # 12 AWG. Conductor material shall be copper unless otherwise noted. Conductor type shall be solid for # 12 and stranded for # 10 and larger.
15. Safety regulations (lock out-tag out, etc.) must be observed by the contractor during construction.
16. The wiring size is based on the estimated conduit routing as shown in this drawing package. Should the conduit's length increase due to relocation of source and/or routing, the conduit and wire may need to be resized. Please contact the Architect/Engineer prior to making any field changes.
17. All copper wiring shall be XHHW or XHHW-2 for 90 C applications, DC Wire shall be 1000V rated PV wire for 1000V rated. All aluminum wiring shall be XHHW for 90 C applications and have compression lugs at terminations. Use bare copper for ground where applicable.
18. Inverter interconnection via a bus tap is only legal when an over current protection (fusible AC disconnect) is located within 10 feet of the tapped conductors. Per NEC Article 240, the conductors shall be crimped with a single hole crimp-on lug, manufactured by ILSCO or BURNDY. Lugs shall be constructed of pure copper and tin plated for high conductivity. The lugs must be rated for 600 Volt DC applications minimum. The crimp must be made with the manufacturer's approved device to achieve the proper crimp connection. Use stainless steel or other non-corrosive hardware with the fastener torque to manufacturer's recommendations on all three phases. This torque level is a requirement. Minimum bend radius shall be observed to maintain good conductor quality and wire management in the load center or transformer. If this bend radius is too constricting, use a 90 crimp offered by ISLCO or BURNDY. Ensure that acceptable clearances for safe continuous operation are allotted with bus tap.

19. Conductor and conduit sizing calculated on continuous duty and MAX 2% voltage drop where possible.
20. All conduits shall be free of any obstructions before wire is pulled.
21. Electrical contractor to provide signage as shown on these drawings and per NEC Article 690.
22. Unless otherwise indicated, ground all exposed noncurrent-carrying metallic parts of electrical equipment, raceway systems, equipment structures and the neutral of all wiring systems in accordance with the NEC, State, and other applicable laws and regulations.
23. Where ground rods are indicated or used, they shall be copper clad, not less than 3/4" in diameter, and 8 feet long and driven full length into the earth. Make ground connections by brazing, exothermic welding, or with approved pressure terminals or mechanical grounding devices, except inaccessible connections shall be made by exothermic welding. The point of contact of each exothermic weld shall be wire brushed or filed to a bare metal surface. Thermite welding cartridges and molds shall be used in accordance with the manufacturer's recommendations. After the welds have been made and cooled, slag shall be brushed from the welded area and the joint thoroughly cleaned.
24. Trenches shall not be left unattended unless the area is fenced or barricaded to restrict entry to the area. Call the underground service firm before trenching. Warning tape shall be laid in trenches at a depth of 12 inches below finished grade and at least 12 inches above installed conduit, it shall be laid on the compacted backfill for the full length of the trench. Do not stretch the tape; installation of warning tape under slabs is not required. Conduit trenches shall be free of rocks, debris, etc. The trench shall be inspected by owner/Engineer or representative thereof prior to conduit installation.
25. Mini power centers (when included) shall consist of a combination dry type transformer with primary breaker, a secondary breaker, and a panel board. Breakers shall be plug-on type. Transformer shall have 115 degree C rise insulation. The complete unit shall be UL labeled and be suitable for outdoor use. Provide Square D mini power-zone, Cutler-Hammer mini-power centers, or equal.
26. Receptacles shall be duplex and rated 20 Amps at 120 VAC, 2 Pole, 3 Wire, NEMA type 5-20R and specification grade. All receptacles shall be GFI and weatherproof type. Device cover plates shall be suitable for the environment in which they are installed and the type of service they are used for.
27. Heavy duty rated switches fused or non-fusible as indicated on the drawings, shall be provided as required. General duty switches will not be allowed. Switches shall have "quick-break" actuating mechanisms and shall be enclosed as required by the conditions of installation. The cover shall be interlocked with the switch such that the enclosure cannot be opened with the switch in the "on" position. The "on" and "off" positions shall be clearly marked by the manufacturer. The switch shall be capable of being locked in the open position. Provide enclosures suitable for the specific type of location in which they are installed. Provide visible blade switches were required by code or utility. Disconnect switches shall be manufactured by Cutler-Hammer, Square D or equal.
28. Provide detectable underground warning tape at all feeders on primary of GSU transformer.
29. E.C. to provide red-lined as-builts at the completion of the project. A red lined set shall be maintained and accessible on site throughout construction.
30. Contractor shall be required to review conduit and wiring routing to prevent moisture from entering combiner boxes. Provide submittals to owner and owner's representative for approval, of all wiring, combiner boxes, conduit and other major BOS components.
31. All exposed conduit shall be installed to accommodate expansion and contraction due to ambient temperature changes per NEC requirements.

Module Mount Racking

1. Racking will consist of the specified manufacture as noted in layout & details.
2. Provide tilt angle per the project drawings.
3. Provide rack manufacturer's ballast and anchorage calculations and shop drawings to Architect / Engineer for review and project confirmation prior to installation. (When applicable)
4. Verify rack mounting rail spacing with module manufacturer's support requirements. Report any discrepancy immediately to Architect / Engineer for clarification prior to installation of any PV modules.

Inverter Installation Notes

1. Inverters shall be installed and wired per the manufacturer's installation manual.
2. Verify inverter output voltage rating equals the utility line voltage at the point of connection.

Monitoring and Data Acquisition

1. Monitoring system shall include data logger, weather station, revenue meter and sensors as shown in these drawings.
2. Monitoring system components shall be installed per manufacturer's instructions.
3. Wiring from components to equipment and to control panels must be continuous with no splices.

Solar Array Commissioning

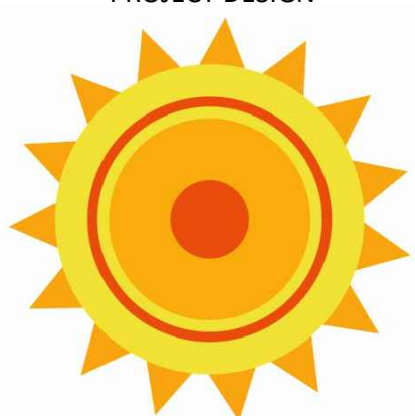
1. Before closing disconnects or attempting to energize the inverters, the following commissioning procedure shall be completed:

A. Contractor to follow system owner approved commissioning procedure per PV Technical Specification.

String Combiner Boxes

1. Box shall have individual fuses for each string.
2. Provide NEMA 3R enclosure. (Minimum)
Array Combiner Boxes (When required)
 1. Provide multiple input combiner box with individual fuses per single line diagram.
 2. Provide MIN. NEMA 3R enclosure.
 3. Provide remote monitoring. (when specified)

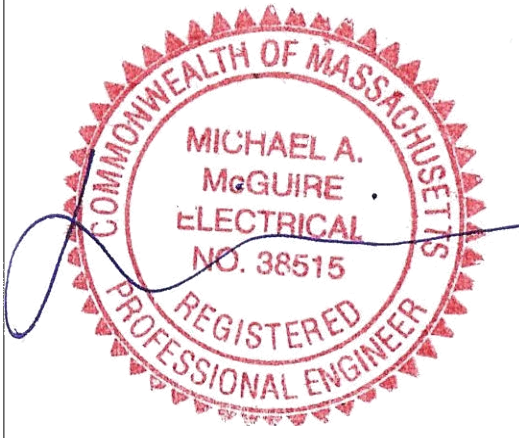
PROJECT DESIGN



SURYA
Standard
Harnessing the best source of energy available

3 Allied Drive, Suite 107
Dedham, MA 02026
amanullah@suryastandard.com
(617) 842-8016

SEAL/SIGNATURE



sealed 03aug2026 mike@h2dc.com
H2DC PLLC MA CoA#: 001239603
ELECTRICAL ONLY
FOR PERMITTING PURPOSES ONLY

PROJECT

**BTM ROOFTOP
PHOTOVOLTAIIC SYSTEM**
DWARKA LLC
86 SANDERSON AVENUE, LYNN, MA 01902

PROJECT OWNER
DWARKA LLC

PROJECT DEVELOPER



SURYA
Standard
Harnessing the best source of energy available

3 Allied Drive, Suite 107
Dedham, MA 02026
amanullah@suryastandard.com
(617) 842-8016

REVISIONS		
DATE	COMMENT	REV.
5/11/2026	INTERCONNECTION PLAN SET	1.0
8/3/2026	PERMIT PLAN SET	1.0

PROJECT NO:

FILE NAME: DWARKA LLC

DRAWN BY: SPV

CHECKED BY:

COPYRIGHT:

PERMIT PLAN SET

SHEET TITLE
**GENERAL AND
ELECTRICAL NOTES**

SHEET NUMBER
E-207